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Wrzus, C., & Roberts, B. W. (2017). Processes of Personality Development in Adulthood: The TESSERA Framework. *Personality and Social Psychology Review, 21*, 253-277. doi:10.1177/1088868316652279

Processes of Personality Development in Adulthood: The TESSERA Framework

Cornelia Wrzus[1] and **Brent W. Roberts**[2]

[1]Johannes Gutenberg University, Mainz Germany, [2] University of Illinois, Urbana-Champaign, USA

The current article presents a theoretical framework of the short- and long-term processes underlying personality development throughout adulthood. The newly developed TESSERA framework posits that long-term personality development occurs due to repeated short-term, situational processes. These short-term processes can be generalized as recursive sequence of Triggering situations, Expectancy, States/State expressions, and Reactions (TESSERA). Reflective and associative processes on TESSERA sequences can lead to personality development (i.e., continuity and lasting changes in explicit and implicit personality characteristics and behavioral patterns). We illustrate how the TESSERA framework facilitates a more comprehensive understanding of normative and differential personality development at various ages during the lifespan. The TESSERA framework extends previous theories by explicitly linking short- and long-term processes of personality development, by addressing different manifestations of personality, and by being applicable to different personality characteristics, e.g., behavioral traits, motivational orientations, or life narratives.

Keywords: long-term personality development; short-term processes; lifespan; social-cognitive learning; person-situation transaction

Author note. The authors thank Mitja Back, Anna Baumert, Brent Donnellan, Boris Egloff, Jeanette Krohn, Marcus Mund, Martin Quintus, Elliott Tucker-Drob, Jenny Wagner, and the participants of the 2014 SPSP pre-conference “Lifespan personality psychology” for their valuable comments on ideas and drafts of this paper. Correspondence concerning this article should be addressed to Cornelia Wrzus, Psychological Institute, Johannes Gutenberg University Mainz, 55122 Mainz, Germany. E-mail: wrzus@uni-mainz.de.

People's personality shows continuity and at the same time it changes throughout adulthood, often in relation to specific life experiences and transitions (Lodi-Smith & Roberts, 2007; Lüdtke et al., 2011; Roberts & DelVecchio, 2000; Roberts, Walton, & Viechtbauer, 2006). For example, emotional stability increased over the course of four years when young adults entered a romantic relationship for the first time, but not when they remained single (Lehnart, Neyer & Eccles, 2010; Neyer & Asendorpf, 2001). The prevailing explanations of such patterns of development often refer to experiences in daily life that evoke long-term change in personality over time. Yet, details about the nature of such daily experiences, the link to long-term personality development, and reasons why some experiences predict change, whereas others predict continuity have not been specified.

The goal of this paper is to identify the basic parts and processes[1] of daily experience and behavior leading to patterns of continuity and change in personality over time (i.e., personality development). In the TESSERA framework, these parts are recursive sequences of **T**riggering situations, **E**xpectancies, **S**tates/**S**tate **e**xpressions, and **Rea**ctions (TESSERA). Thus, multiple specific experiences add to a person's characteristics like tesserae, little stone pieces, to a mosaic. The general proposition that personality development occurs as a result of daily experiences that cumulate over time has recently gained attention (Back et al., 2011; Hennecke, Bleidorn, Denissen, & Wood, 2014; Roberts, 2006, 2009). Yet, previous work largely focused primarily on personality states, and did not detail the processes through which repeated specific experiences contribute to the development of personality characteristics measurable with self-reports, other-reports, or behavioral observation.

Research on the processes of personality development has been called for repeatedly (Durbin & Hicks, 2014; Hopwood et al., 2009). Especially a comprehensive framework on the linkages of short- and long-term processes in personality development in adulthood is currently lacking. Such a framework could guide empirical work on better understanding long-term personality development, and provide a comprehensive theoretical background for interventions that already focus on daily experiences to elicit long-term change, such as during clinical therapy (De Fruyt, Van Leeuwen, Bagby, Rolland, & Rouillon, 2006).

In the first section, we briefly summarize the patterns of personality continuity and change from young adulthood to old age. In the second section, we propose a framework for the short- and long-term processes of personality development that incorporates and expands previously proposed developmental and personality processes. Here, we extend social-cognitive behavioral theories (e.g., PERSOC Back et al., 2011; CAPS Mischel & Shoda, 1995; AMP Krampen, 1988) and explicitly link the assumed short-term processes to long-term personality development. Third, we discuss how the proposed framework facilitates the understanding of different patterns of personality development throughout adulthood. Fourth, we address the application of the framework to different personality domains. Finally, we discuss the contribution of the TESSERA framework for understanding personality development, provide unique predictions, and outline future research directions.

Previous Knowledge on Personality Development in Adulthood

As a starting point, we view our process model of personality development as embedded in comprehensive conceptualizations of personality (McAdams & Pals, 2006; McCrae & Costa Jr, 2008; Roberts & Wood, 2006). Accordingly, personality constitutes characteristics that reflect relatively enduring patterns of typical cognition, emotion, motivation, and behavior in which individuals differ from others of the same culture or subpopulation (e.g., DeYoung, 2015; Kandler, Zimmermann, & McAdams, 2014; McCrae & Costa, 2008). Such enduring patterns include general

behavioral dispositions (e.g., Big Five traits), motivational orientations (e.g., goals, values, attitudes; DeYoung, 2015; Kandler et al., 2014; McCrae & Costa Jr, 2008), as well as life narratives (i.e., the subjective, integrative "story" of a person's characteristics and experiences; McAdams & Pals, 2006; McAdams & Olson, 2010; Roberts & Wood, 2006). Personality characteristics manifest and can be measured on different levels: explicit mental representations, implicit associative representations, biological functioning, observable behavior, and reputations, which other people report. For the sake of clarity, we initially focus on explicit, implicit and behavioral manifestations of dispositions (e.g., Big Five traits) as an example to describe the framework of short- and long-term processes of personality development. In the final sections of this paper, we elaborate that the framework applies equally well to other personality domains, such as goals, self-esteem and narrative descriptions and discuss possible future research directions concerning reputations and biological manifestations.

Nomothetic (variable-oriented) personality development encompasses continuity in relative standing, such as documented in the rank-order stability of personality, as well as persistent, differential changes in absolute mean-level (increases or decreases) over time (Caspi & Roberts, 1999; Roberts & Caspi, 2003). The rank-order stability of, for example, Big Five traits increases from adolescence to adulthood up to r = .60—.70 and plateaus (Roberts & DelVecchio, 2000). Recent evidence shows that rank-order stability might decrease somewhat in late adulthood (Ardelt, 2000; Lucas & Donnellan, 2011; Wortman, Lucas, & Donellan, 2012). The normative mean-level change of Big Five traits in young and middle adulthood mainly consists of increases in agreeableness, conscientiousness, emotional stability, and social dominance (Lucas & Donnellan, 2011; Roberts & Mroczek, 2008; Roberts et al., 2006b). In old age, several studies show a reversed pattern with longitudinal decreases in agreeableness, conscientiousness, emotional stability, and openness (Berg & Johansson, 2014; Kandler, Kornadt, Hagemeyer, & Neyer, 2015; Lucas & Donnellan, 2011; Mõttus, Johnson, Starr, & Deary, 2012b).

Such trait changes often occur together with life transitions (Bleidorn, 2012, 2015; Le, Donnellan, & Conger, 2014; Lodi-Smith & Roberts, 2012; Specht, Egloff, & Schmukle, 2011; Zimmermann & Neyer, 2013), and experiences in personal relationships (Mund & Neyer, 2014; for reviews see Neyer, Mund, Zimmermann, & Wrzus, 2014; Wrzus & Neyer, 2016) as well as work experiences (Roberts, Caspi, & Moffitt, 2003; Hudson, Roberts, & Lodi-Smith, 2012). Although several theories address long-term personality development (see Specht et al., 2014 for a comparison of theoretical accounts of personality development), little is known about the underlying short-term processes that lead to long-term continuity and change (Durbin & Hicks, 2014; Roberts, 2009; Specht et al., 2014). We next offer a description of the TESSERA framework of personality development that links short- and long-term processes.[2] Here we draw on previous work from different research traditions such as personality, social, developmental, and clinical psychology.

TESSERA: A Framework of Personality Development Processes

So far, empirical investigations have mainly studied personality trait change over several years and related those changes to single life events, such as new partnership, divorce, or first employment, that had occurred in the intervening years between the typical assessments of personality (Lüdtke et al., 2011; Neyer & Lehnart, 2007; Orth, Robins, & Meier, 2009; Specht et al., 2011). Such investigations sometimes acknowledge that personality change occurs gradually over time, yet the exact short-term processes of how life events translate into repeated daily situations and experiences, which eventually change behavioral patterns and personality representations, are not examined. We

postulate that repeated short-term sequences of Triggering situations, Expectancies, States/ State expressions, and Reactions (TESSERA) are the puzzle pieces, i.e., tesserae, and *reflective* and *associative processes* transfer TESSERA sequences into long-term personality development (Figure 1, Table 1).

The Short-term Components of the TESSERA Framework for Personality Development

One of the key distinctions necessary for making the link between short-term experience and long-term change is to identify processes, which go beyond a single event that perturbs a person's behavior to a series of events that change a pattern of behavior over time. A recent study (Penton-Voak et al., 2013) provides a provocative example and can be used to better understand the components of the TESSERA framework. Specifically, in their study, aggressive adolescents viewed ambiguous faces on computers [Trigger], perceived and rated the faces as angry or not [State thought and State expression], and received feedback whether the faces were indeed angry [Reaction].[3] Initially, aggressive adolescents tended to view ambiguous faces as more angry than friendly. After repeated positive feedback when classifying ambiguous faces as more friendly in the experimental group, adolescents' level of aggressiveness decreased over two weeks as indicated by daily self-reports and observed aggressive behavior in daily life (Penton-Voak et al., 2013). Clearly, longer follow-up assessments are needed to examine the psychological processes leading from training to changes in self-descriptions and behavior and lasting changes in trait aggressiveness. Still, the initial changes support the idea behind the TESSERA framework, which is that the accumulation of state change would eventually lead to personality change observable to oneself and others. Now we briefly describe each component of the TESSERA framework (Table 1), relate them to previous research (e.g., PERSOC Back et al., 2009, 2011; CAPS Mischel & Shoda, 1995), and highlight how the components contribute to long-term personality development.

Triggering situation. Events or daily situations external to the person and occurring in multiple contexts can serve as triggers. Previous research discussed as contexts for development: (a) (life) events that include several experiences (e.g., divorces including conflicts; Filipp, 1990; Holmes & Rahe, 1967); (b) social roles that assemble specific experiences (e.g., supervisors encountering leadership situations; Roberts, 2006; Roberts et al., 2003); (c) interventions or therapies (De Fruyt et al., 2006, 2012; Magidson et al., 2014; Webb & Sheeran, 2006); (d) others' behavior or words (model learning; Bandura, 2003). Thoughts, such as about goals to change personality (Hennecke et al., 2014; Hudson & Roberts, 2014; McAdams, Ruetzel, & Foley, 1986; Snyder & Ickes, 1985), are discussed under *Environmental and Personal Factors Contributing to TESSERA Sequences.* Within the TESSERA framework, single situations trigger expectancies and/or states (Figure 1). Thus, when studying personality development associated with larger (life) events, for example marriage, researchers need to zoom in to find out what exactly are the relevant triggering situations (e.g., conflicts in daily interactions and their solution).

The conceptualization and assessment of situations bear some challenges relevant for examining personality development. Situations can be characterized by their physical (e.g., location, activity, persons) and/or psychological properties (e.g., task-related, threatening, pleasant; for reviews see Rauthmann, 2015; Rauthmann, Sherman & Funder, 2015; Wagerman & Funder, 2009). Although, physical properties can directly elicit states (e.g., emotions, habitual behavior, Wood & Neal, 2007), psychological properties seem to predict states better (e.g., Fleeson & Jolley, 2006; Sherman et al., 2015; Mischel & Shoda, 1995; Rauthmann et al., 2015). Second, distinguishing situations seems necessary to understand why some situations, but not

others, elicit trait-relevant states. Most researchers concur that situations differ when the physical (Magnusson, 1981) or the psychological environment changed (Rauthmann, 2015; Rauthmann, Sherman, Nave, & Funder, 2015). At the same time, the subjective novelty of a situation might deviate from more objective information. Situations then differ when they cannot be incorporated in existing schemata (broadly related to Block, 1982; Piaget & Inhelder, 1969). The TESSERA framework postulates that triggering situations and their perceived psychological meaning (including novelty) are necessary but not sufficient parts of personality change, which will not occur without trait-relevant states (see Figure 2). Situations can trigger expectancies of how to behave, feel, or think in that situation (Wood & Denissen, 2015);); alternatively, situations can trigger states directly (e.g., if-then contingencies Mischel & Shoda, 1995, 1998; Rauthmann et al., 2014; Schmitt et al., 2013; Figure 1).

Expectancy. Expectancy refers to the momentary motivational construct that guides which state (i.e., thoughts, feelings, and behavior) occurs after the trigger by "selecting" a response from the variety of possible states (Fleeson & Jayawickreme, 2015; Heckhausen & Gollwitzer, 1987; Wood et al., 2015; Wood & Neal, 2007; Wood & Roberts, 2006). Thus, the connection between triggering situation and state is not only an "automatic" stimulus-response sequence as behaviorism would propose (Watson, 1913) but can also occur intentionally. Intentions and goals are thought to contribute powerfully to behavior and behavioral change (Aarts & Dijksterhuis, 2000; Bandura, 1977; Durbin & Hicks, 2014; Heckhausen & Gollwitzer, 1987; Heckhausen, Wrosch, & Schulz, 2010; McCabe & Fleeson, 2016; Mischel & Shoda, 1995; Webb & Sheeran, 2006; Wood & Denissen, 2015). Often situations include own and others expectancies (see *Moderators of the TESSERA Components* for distinction) about the possible, appropriate or useful behavior, or state more generally. Accordingly, appropriate or useful behavior means that the behavior serves to achieve a goal. For example, being in an *employee* role might repeatedly elicit disciplined and orderly behavior, partly due to expectations about role-appropriate behavior that serves the goal to be a dependable and successful employee (Lodi-Smith & Roberts, 2007; Roberts & Wood, 2006).

Health-behavior models further elaborate the role of expectancies for behavioral change (e.g., the transtheoretical model of change; Marschall & Biddle, 2001; Nigg et al., 1999; Prochaska & DiClemente, 1982). Expectancies guide the behavior towards the anticipated goal, such as quitting smoking or starting exercising, yet they may not always be necessary (Figure 1, see also Wood & Neal, 2007). For example, a meta-analysis on health-related behavioral interventions (Webb & Sheeran, 2006) showed that behavior also changed when interventions elicited no change in intentions. Still behavioral change was somewhat larger when interventions increased behavioral intentions, which then elicited behavior change.

State and state expression. Momentary thoughts, feelings, and behavior[4], are the core aspect of personality and personality development as personality characteristics can be defined as individual differences in repeated, relatively time- and situation-consistent behaviors, thoughts, and feelings (Fleeson & Jayawickreme, 2015; Fleeson & Jolley, 2006; Hooker & McAdams, 2003b; Magidson et al., 2014; Mischel & Shoda, 1995; Roberts & Jackson, 2008; Wood & Neal, 2007). Personality trait change presumably comes about because repeated states get ingrained and become long-term, consistent patterns (Fleeson & Jayawickreme, 2015; Magidson et al., 2014; Roberts, 2009; Wood & Neal, 2007), the processes are detailed in the section *Processes of Transferring Short-term TESSERA Sequences into Long-term Personality Development*.

Empirical evidence for the linkage between state change and trait change is beginning to accumulate. First empirical studies showed that repeated experiences of

increased momentary negative affect in daily stress situations predicted a longitudinal increase in neuroticism over six years (Wrzus, Luong, Wagner, & Riediger, 2016). We assume that only repeated states that differ from one's typical, trait-like behavior can elicit personality change (Nesselroade & Molenaar, 2010; Roberts & Jackson, 2008). That means a person would have to be more stressed than usual (i.e., increased negative affect after hassles) to explain long-term changes in neuroticism. Otherwise existing trait levels are strengthened leading to personality continuity (see Figure 2).

Behaviors, but also thoughts and feelings, nearly always elicit reactions—either in other individuals (e.g., the partner sulks after being yelled at) or in oneself (e.g., feeling bad after thinking spitefully about the partner). Before other individuals can recognize and react upon thoughts and feelings, they have to be expressed in overt behavior (e.g., words, facial expressions, actions; Back et al., 2011a; Mischel & Shoda, 1995).

Reaction. Reactions can come from oneself (reinforcement through emotions) or others (reinforcement and punishment, e.g., verbal feedback, facial behavior such as smiling or frowning) in response to states. Others' reactions reaction need to be perceived to function as reinforcement for individual (e.g., Back et al., 2011a). Reinforcement learning through continuous approach toward positive feelings and avoidance of negative feelings is a strong learning mechanism for a multitude of domains including complex behaviors and personality (Caspi & Roberts, 2001; Gerlsma, Emmelkamp, & Arrindell, 1990; Kanfer & Grimm, 1977; Kanfer & Phillips, 1970; Mischel & Shoda, 1995; Roberts & Wood, 2006). Punishment conveys the demand that certain behaviors, thoughts, or feelings need to change and future rewards or punishments would follow if appropriate changes in behavior are or are not made. When such reactions are incongruent with their personality characteristics, individuals are presumably more likely to change (Burke, 1991; Caspi & Roberts, 2001). Thus, reactions perpetuate TESSERA sequences and thus provide one link to how single sequences persist or change over time. Empirical evidence for this type of process is still lacking as no study to our knowledge has tracked the effects of continuous reactions to specific states on personality development.

Repetition of TESSERA sequences. We emphasized before that the crucial factor to link short-term experiences to long-term change is repetition (as indicated by the plus signs in Figure 1). One way of how repetition of similar TESSERA sequences occurs is that previous TESSERA sequences render future similar TESSERA sequences more likely, for example, through reactions or reinforcement learning. In addition, states after specific situations may elicit or create similar situations later on: Feeling anxious increases the likelihood of experiencing or attending to threatening triggers subsequently (MacLeod, Rutherford, Campbell, Ebsworthy, & Holker, 2002), which is discussed as a cycle of developing anxiety disorder. Other individuals also can be "the situation" and reciprocal behavior among two interaction partners, for example during flirting (Back et al., 2011b), can be understood as repeated trigger-state sequences: Person A's flirting behavior elicits flirting behavior in Person B, which serves as trigger to elicit further flirting behavior in A (see also Back et al., 2011a). Constant demands of the environment and personality dispositions creating or seeking similar triggering situations can also lead to repeated TESSERA sequences.

Environmental and Personal Factors Contributing to TESSERA Sequences

We assume that environmental factors can influence the TESSERA processes in multiple ways (Figure 1). For example, the proximate and cultural environment (i.e., the rather stable physical and social surroundings compared to temporally varying situations) offers and constrains the kind of situations people experience, the expectations people face, the range of the appropriate behaviour, and the external reactions (e.g., reactions by other individuals; Back et al., 2011a,b). As much as environments are relatively constant

in their expectancies (or push) for specific states, repeated similar TESSERA sequences are elicited. For example, many full-time work environments likely demand accuracy and dependability, eliciting behavior related to conscientiousness in most individuals. Yet, environments can also differ within individuals over time and between individuals (e.g., differences in work or residential environments). One study examined residential factors and changes in trait well-being at the end of life (Gerstorf et al., 2010): Decreases in well-being were exacerbated in socio-structurally problematic environments (e.g., low health care, unemployment), which likely provided more negative daily situations. Cross-cultural research further provides evidence on how environments differ in accepted behavior, for example, in reactions to different intensities of extraversion (Ward, Leong, & Low, 2004), which may limit the possible changes in personality characteristics.

In addition, individuals' dispositions (e.g., Big Five traits), goals, abilities, or health[5] may constrain the triggers and states they experience (Figure 1) as individuals select or create personality-congruent situations (Buss, 1987, Roberts & Robins, 2004; Wrzus, Wagner, & Riediger, 2016). Personality characteristics also circumscribe potential states: for example, the trait level of extraversion limits how much and how often a person behaves extraverted in specific situations (Fleeson, 2001; Fleeson & Gallagher, 2009). Similarly, goals to change may contribute to, for example, selecting triggers or forming expectancies (Hennecke et al., 2014; Hudson & Frailey, 2015; Hudson & Roberts, 2014; Martin, Oades, & Caputi, 2015; McAdams, Ruetzel, & Foley, 1986; Snyder & Ickes, 1985). Finally, internal reactions (e.g., joy or disappointment) partly relate to stable personal factors (e.g., dispositional affect). Personal factors assessed on the level of psychological manifestation (e.g., traits, abilities) influence the processes of personality development (e.g., Wagner, Ram, Smith, & Gerstorf, 2015; Wrzus et al., 2016); yet, personal factors assessed on the level of genetic differences, might additionally influence personality development, perhaps partly mediated through psychological manifestations (see *Future Research Directions*).

Moderators of the TESSERA Components: Valence, External-Internal Locus, and Automaticity

So far, the TESSERA framework is generic. Triggering situations can be unpleasant social interactions or pleasant reading in solitude. The same is true for the other TESSERA components. Within the TESSERA framework this heterogeneity in the components can be described by moderators. Specifically, we propose that TESSERA components may vary on the dimensions of positive—negative valence, external—internal locus, and automatic—deliberate (Table 1). As a consequence, the size or direction of personality development may vary.

Positive—negative valence. All TESSERA components can vary in valence. Regarding triggering situations, research on life transitions shows that (perceived) positivity and negativity of transitions predict diverging patterns of personality trait changes (Lüdtke et al., 2011; Specht et al., 2011). We further assume that differences in valence influence the size of personality development. In general, negative events, states (emotions, behaviors), and reactions have stronger consequences in various domains such as well-being, health, cognition, or interpersonal outcomes (Baumeister, Bratslavsky, Finkenauer, & Vohs, 2001). Accordingly, we assume that negative triggering situations, states, or reactions elicit stronger (not necessarily more favorable) personality development as also predicted from theories about the adaptive[6] value of negative emotions (Carver, 2004). For example, Wrzus et al. (2016) linked increased hassle reactivity in negative affect to longitudinal increases in neuroticism, and we hypothesize that positive affect reactions to pleasant daily events lead to less pronounced changes (i.e., decreases) in neuroticism.

External—internal locus. The TESSERA components can also vary in being external or internal to a person (i.e. internal expectations, thoughts, reactions)—except for triggering situations, which we conceptualize as external. If thoughts about a situation trigger further states, the triggering situation is represented internally, but was external originally. A special case of externality exists, when some of the components are social—outside of a person and related to other persons. For example, the PERSOC framework on the interplay of personality and social relationships (Back et al., 2011a) argues that social interactions lead to long-term changes in personality traits and relationship schemas. The TESSERA framework is in line with this prediction and additionally focuses on the inanimate environment (e.g., when landscapes elicit specific thoughts or feelings) as well as internal expectations and reactions (e.g., when own feelings reinforce and own thoughts provide feedback on a certain behavior).

We predict that externality-internality of TESSERA components moderate the size of personality development. For example, social triggers or reactions should lead to larger changes compared to non-social triggers or reactions because previous research showed that social relationships are powerful catalysts of personality development (Finn, Mitte, & Neyer, 2015; Mund & Neyer, 2013). In contrast, external expectations should result in smaller changes compared to internal expectations because intrinsic motivation facilitates behavior and goal attainment more strongly than extrinsic motivation (Ryan & Deci, 2008; Sheldon, Ryan, Deci, & Kasser, 2004; Sheldon & Schüler, 2011).

Automatic—deliberate. The TESSERA components and their links can occur more or less deliberately and with conscious attention. Vast evidence exists that features of situations can be perceived and can elicit states with little or no conscious attention, expectations can influence behavior without conscious attention, and own and others' reactions can occur automatically (for reviews Bargh & Ferguson, 2000; Deutsch & Strack, 2006; Wilson, 2004; Wood & Neal, 2007). Also, personality characteristics can influence states both more deliberately and more automatically (Back, Schmukle, & Egloff, 2009). Importantly, we distinguish the automaticity of TESSERA components and their links from associative and reflective processes integrating TESSERA sequences into trait development (Figure 1). For example, rather automatically elicited behavior (e.g., bad habits) can be consciously perceived and reflected—altering a person's explicit self-perception. Also, situations that have been deliberately sought out to experience new levels of states can lead to changes in implicit personality characteristics through implicit learning.

Some recent theoretical work explicitly focuses on deliberate personality development and highlights the active role of the individual in achieving personality continuity and change (Hennecke et al., 2014; Hudson & Roberts, 2014; Wood & Roberts, 2006;). Currently, we can only speculate that personality development is more pronounced when TESSERA processes occur more deliberately. For example, two intervention studies showed that the more individuals strived to change personality characteristics, the more their self-perceived traits changed over the next few months (e.g., Hudson & Fraley, 2015; Martin, Oades, & Caputi, 2014).

Processes of Transferring Short-term TESSERA Sequences into Long-term Personality Development

To identify processes of personality development we conducted a literature search (PsycINFO, ScienceDirect) using the keywords *personality development*, *identity development*, *attitude change*, or *behavioral change* as well as using backward and forward citations of central theoretical work (e.g., Caspi & Roberts, 2001; McAdams & Olson, 2010; Roberts & Wood, 2006). We focused on theoretical work and clustered the various processes (Table 2) into the often applied broad classes of (a) reflective processes and (b) associative processes (e.g.,

Back, Schmukle, & Egloff, 2009; Gawronski & Bodenhausen, 2006; Rothmann, Sheeran, & Wood, 2009; Strack & Deutsch, 2004). This distinction, although not necessarily exclusive, allows linking the processes to different manifestations of personality characteristics (e.g., explicit or propositional, implicit or associative, and behavioral personality manifestations; reputational and biological manifestations of personality are discussed in the section on *Future Research Directions*). When discussing associative and reflective processes next (see also Table 1), we point out similarities among the processes, which are summarized and ordered by time scale in Table 2.

Associative processes. Repeatedly experiencing TESSERA sequences can lead to changes in implicit personality characteristics through implicit learning processes (Seger, 1994; Table 2). More specifically, repeated behavioral states, trigger-behavior links, or goal-action patterns are encoded in the procedural memory as habitual behavior (i.e., habits; Aarts & Dijksterhuis, 2000; Wood & Neal, 2007; Table 2). In addition, states, which are activated repeatedly and together with the concept *self*, are encoded as implicit associations in the associative memory (Back et al., 2009; Back & Nestler, in press) and become stronger and more accessible with more repetition (Higgins, 1996). For example, repeated attention to negative triggers and consequently experience of intense negative affect is discussed as a pathway for trait anxiety and anxiety disorder (MacLeod et al., 2002; Matthews & Mcleod, 2002, 2005; See et al., 2009). In reversal, repeated attention on pleasant triggers, for instance during cognitive bias modification training, reduces trait anxiety (Hallion & Ruscio, 2011).

Reinforcement learning can add to the implicit learning if states are repeatedly associated with pleasant or unpleasant reactions (e.g., own feeling, other's verbal or non-verbal reactions). Thus, implicit learning, reinforcement learning, and habit formation are associative processes capable of explaining personality development on the level of implicit representations and behavioral pattern (Table 2). In contrast to implicit learning, reinforcement learning emphasizes the importance of pleasant and unpleasant reactions. Habit formation can result from both implicit and reinforcement learning but focuses on behavior, not cognitive or affective states.

The TESSERA framework does not equate repeated states (i.e., chronic behavior or affect) in response to *persisting* situations with personality development. Personality development needs to manifest on different levels (e.g., implicit self-concept, others' perception based on outward behavior), in behavioral domains broader than single behaviors, and across different contexts to be considered personality development. In our initial example of training aggressive adolescents to perceive ambiguous faces as less angry (Penton-Voak et al., 2013), changes in the training group were observed in self-rated and other-rated aggressive behavior in daily life. Thus, the trained state (anger perception of faces) generalized to other states (e.g., verbal and physical aggression), across contexts and across assessments of personality. Presumably, learning to interpret ambiguous faces as neutral (instead of hostile), created new, and relatively enduring cognitive schemata for neutral faces. Accordingly, neutrally perceived faces triggered less perceived threat, which reduced spontaneously shown aggressive behavior and the self-perception as aggressive (Penton-Voak et al., 2013). The new schemata could then facilitate future personality development by seeking or creating environments that reinforce the original personality trait (see *Corresponsive principle*, Caspi & Roberts, 2001; Roberts & Wood, 2006; for similar arguments on cognitive traits see Dickens & Flynn, 2001; Durbin & Hicks, 2014).

Reflective processes. Reflective processes presumably change and maintain personality by consciously thinking about one's past experiences, behavior, thoughts, and feelings. Such processes might take place, for example, as part of volitional development (i.e., people as agents of their

own development, Greve, Rothermund, & Wentura, 2005; Hennecke et al., 2014; McAdams, 2015) or therapeutic interventions (Tang & DeRubeis, 1999). Remembering, (re)structuring, evaluating, and reappraising experiences are integral parts of reflective processes such as life reflection (Staudinger, 2001), self-narration (Datan, Rodeheaver, & Hughes, 1987; Hooker & McAdams, 2003a; King, 2001; McLean, Pasupathi, & Pals, 2007; Pals, 2006), self-reflection (Bem, 1972; Caspi & Roberts, 2001; Robins & John, 1997; Wilson & Dunn, 2004), accommodation, and assimilation (Brandtstädter, 1989), or specific therapeutic interventions (Mayer, 2004, Table 1). These reflective processes differ in their scope. Reflections can cover very short experiences, such as thinking about a specific stressful interaction or positive experience with one's spouse, or longer, multiple experiences, such as the progress, the conflicts, and pleasant experiences of the first months of one's marriage (McLean et al., 2007; Pals, 2006). Life reflections and self-narratives usually cover the time lived so far and most important events (Hooker & McAdams, 2003b; King, 2001; Staudinger, 2001; Table 2).

We propose that single TESSERA sequences are either immediately reflected upon (i.e., remembered, restructured, evaluated) or are recalled later when something or someone triggers to form an explicit account of one's personality characteristics. For example, when individuals report difficult experiences (i.e., life narratives) they often follow the structure of triggering situations, states, and reactions[7], although the complexity, coherence, and amount of reflection differ (McAdams, 1995; Pals, 2006). Individuals then integrate the remembered and evaluated TESSERA sequences into the explicit personality characteristics through confirming or adjusting the propositional (i.e., explicit) representation of personality characteristics (Back et al., 2009; Bem, 1972; Gawronski & Bodenhausen, 2006; Krampen, 1988; Figure 1). Depending on what component of the TESSERA sequence is focused on during reflection, generalized views of self-efficiency (Krampen, 1988), attitudes (Gawronski & Bodenhausen, 2006), behavioral dispositions (Back et al., 2009), or self-esteem form (Hutteman, Nestler, Wagner, Egloff, & Back, 2014c).

During reflection of TESSERA sequences, the episode is remembered and weighed, the components are restructured and evaluated (Flückiger, Grosse Holtforth, Del Re, & Lutz, 2013; Mayer, 2004; Staudinger, 2001). Theoretical models posit that self-perceptions of trait levels should largely track actual rates of trait-indicative states (e.g., Buss & Craik, 1983; Fleeson & Gallagher, 2009). In addition to achieving accuracy, however, aiming at consistency, self-enhancement, or popularity among others can also govern self-perceptions (Robins & John, 1997; Wilson & Dunn, 2004).

When aiming at achieving accuracy, self-perceptions of state and trait level agree with perceptions of other (accurate) observers. In the case of repeated discrepancy, individuals adjust the propositional trait level to the perceived state level (compare to accommodative processes; Brandtstädter, 1989; Piaget & Inhelder, 1969). When individuals aim at achieving consistency, they (re)interpret or assimilate (Piaget & Inhelder, 1969; Robins & John, 1997; Sadler & Woody, 2003) self-perceptions of states in line with existing propositional trait levels. When seeking self-enhancement, individuals (re)interprete self-perceptions of states in line with more positive propositional trait levels than others hold about the person (Kwan, John, Kenny, Bond, & Robins, 2004; Vazire, 2010). When aiming at popularity, individuals perceive and evaluate the state based on the assumed impression they have on other people (Back & Vazire, 2012; Robins & John, 1997). These different aims during self-perception help to explain why repeated state levels do not necessarily translate into matching changes in trait levels of personality characteristics. Changes in self-perceptions should occur if the discrepancy is unpleasant and if the change in the personality characteristic is perceived

as important, desirable, and possible (Brandtstädter & Renner, 1990; see Denissen, van Aken, Penke, & Wood, 2013, Hennecke, et al., 2014 for similar arguments).

Since TESSERA sequences are brief and often rather minor situations, it seems unlikely that single sequences elicit lasting personality change (Figure 2). When single (traumatic) experiences predict changes in personality characteristics, such as reported from war veterans (Jongedijk, Carlier, Schreuder, & Gersons, 1996), it is possible that repeated thoughts of the traumatic experience and the accompanying emotions actually drive personality change (Jayawickreme, & Blackie, 2014).

In sum, we suggest that repeated short-term sequences of triggering situations, expectancies, states/ state expressions, and reactions, lead to personality development through associative and/or reflective processes during repetitions of similar TESSERA sequences (Figure 1). Figure 1 omits recursive paths among the TESSERA components to enhance clarity. Still, we acknowledge recursive paths in the framework. For example, thoughts about a situation might act as triggers and elicit further expectancies and states (Fleeson & Jayawickreme, 2015; Mischel & Shoda, 1995; Lickel et al., 2014) before they receive reactions. Such thoughts presumably occur also during reflection and elicit further internal TESSERA sequences, including states and emotional reactions.

A Note on the Direction and Time Course of Personality Development Processes

The prevailing theoretical position is that most people strive for personality development considered positive (Greve, Rothermund, & Wentura, 2005; Hennecke et al., 2014). Accordingly, studies on ideal future selves attest people's desire to change in normatively valued directions (e.g., Markus & Nurius, 1986; Williams & Gilovich, 2008). In addition, the first studies on personality trait change *motivation* confirm that on average young adults want to increase normatively valued behavioral dispositions (e.g., conscientiousness, emotional stability, Hudson & Roberts, 2014; Hudson & Fraley, 2015). Research on normative personality maturation in young adulthood supports the notion that many people change into the desired direction (Bleidorn, 2015; Roberts, et al., 2008). These developmental patterns are likely linked to successfully accomplishing normative life transitions such as committing to a stable romantic relationship, assuming parental responsibilities, or engaging in work responsibilities (Hutteman et al. 2014a; Lodi-Smith & Roberts, 2007; Neyer & Asendorpf, 2001), and the TESSERA framework offers short- and long-term processes to explain such normative development.

The TESSERA framework is also capable of explaining the absence of change or changes considered maladaptive. Personality changes are absent compared to same-aged people when life transitions and associated triggering situations do not occur (Figure 2); for example, when young adults do not engage in committed partnerships (Neyer & Asendorpf, 2001; Wagner, Becker, Trautwein, & Lüdtke 2015) or regular employment (Lüdtke et al., 2011; Lodi-Smith & Roberts, 2007). In addition, no changes in a certain personality characteristic should occur if the situation does not elicit states relevant for the characteristic or if the state level corresponds to the trait level (Figure 2). For example, stressful situations in daily life repeatedly predicted momentary increases in negative affect, which predicted changes in neuroticism, however changes in other Big Five traits were not consistently predicted (Wrzus et al., 2016). Also, processes of maintaining one's self-view or identity might not lead to changes in individuals' explicit self-representations even when life transitions occur (Brandtstädter & Greve, 1994; Roberts & Caspi, 2003). In such cases, new experiences are (re)interpreted (or assimilated, Piaget & Inhelder, 1969; Robins & John, 1997) in line with existing self-views (Figure 2). Together, these predictions suggest that many steps are necessary for personality change to occur, whereas many "exit points" exist which preclude change

(Figure 2; see also Roberts, 2006; Roberts et al., 2008). These differences in paths leading to change and paths precluding change might explain why personality characteristics are often quite stable (i.e., do not change more) and why it is difficult to change characteristics, for instance, motivated by goals to change and during therapy.

Changes considered maladaptive, such as decreases in emotional stability or extraversion, often occur together with life events mostly experienced as unpleasant (e.g., work or financial problems, severe health issues, or depression; Chow & Roberts, 2014; Lüdtke et al., 2011). The unpleasant life events likely contain specific triggering situations for negative states (e.g., Roberts et al, 2006a) or prevent positive states from happening (e.g., enjoying pleasant interactions with friends). In addition, the same life transition can lead to different directions of personality development depending on the valence of the transition. For example, during the transition to parenthood maladaptive personality development occurred (decreases in conscientiousness and emotional stability) when being a parent was stressful and frustrating, whereas positive personality development occurred (increases in conscientiousness and agreeableness) when feeling competent as parent (Hutteman et al., 2014a; Paris & Helson, 2002).

Is it possible that implicit and explicit personality representations change in different direction? In theory, associative and reflective processes can be separated and should influence implicit and explicit personality characteristics distinctively (Figure 1). Findings on implicit and explicit attitude change confirm this position (Gawronski & Bodenhausen, 2006). The associative-propositional evaluation model (APE model) details conditions under which implicit and explicit attitudes change in the same direction or independently from each other (APE model; Gawronski & Bodenhausen, 2006). Both unidirectional and separate change is also plausible for other personality concepts as the rich literature on discrepancies in implicit and explicit self-esteem, motives, and self-concepts indicates (Asendorpf, Banse, & Mücke, 2002; Briñol, Petty, & Wheeler, 2006; Schröder-Abé, Rudolph, & Schütz, 2007; Spalding & Harding, 1999). Often, however, associative and reflective processes work in concert (Egloff et al., 2008). The usually small association between implicit and explicit personality measures (Hofmann, Gawronski, Gschwendner, Le, & Schmitt, 2005) can be enhanced when information about trait-relevant experiences are actively retrieved (Egloff et al., 2008; Gawronski & Bodenhausen, 2006).

The time course of cumulative change in explicit personality characteristics is theorized to include non-linear changes such as logistic, discontinuous, and reversible patterns (e.g., Durbin, Hicks, Blonigen, & Johnson, 2016; Krampen, 1988; Luhmann, Orth, Specht, Kandler, & Lucas, 2014). First evidence from psychotherapeutic research on depression shows that about one third of clients experience *sudden gains*, that is, notable decreases in depressive symptoms from one session to the next (Lutz et al., 2013; Tang & DeRubeis, 1999). Such sudden gains often increase further during the therapy and persist in follow-up assessments several months and years later (Tang & DeRubeis, 1999). Still the reasons for sudden gains and possible relations to reflective processes are hardly understood. At the same time, relapses are also common in therapy and very plausible regarding non-clinical levels of neuroticism and other personality characteristics. However, such discontinuous patterns of personality development are rarely seen because relatively many and dense personality assessments would be necessary to detect these patterns. For example, changes in subjective well-being related to marriages and widowhood showed curvilinear patterns that varied between individuals depending on their initial change rate (Bisconti, Bergeman, & Boker, 2004; Lucas et al., 2003; Luhmann et al., 2012).

Associative processes, in contrast, should lead to incremental changes in implicit personality representations because associations built up continuously when the

same concepts are activated repeatedly. Only to the extent that inconsistent activation occurs would one expect to find non-linear trajectories. Regarding behavioral patterns, changes in behavior might seem like sudden changes. However, sudden changes in behavior might result from volitional personality processes contributing to behavioral acts, since both impulsive and reflective personality processes influence behavior (Back et al., 2009; Back & Nestler, in press).

Lifespan Personality Development viewed from within the TESSERA Framework

So far, we described the general framework of personality development processes and referred to previous work, which has focused disproportionately on young adulthood. Next, we examine how the TESSERA framework can foster the understanding of personality development across the adult lifespan[8] (Table 1). Specifically, we address three broad developmental patterns: increasing maturity during young adulthood, cumulating continuity during middle adulthood, and diversification or differentiation of change during late adulthood (Table 3). We highlight that late adulthood might be a particularly interesting period, which so far has been rarely addressed in personality research.

Increasing Maturity During Young Adulthood

Across young adulthood, personality mean-levels develop towards greater maturity and regulatory resiliency (Roberts & Mroczek, 2008; Roberts et al., 2006b; Staudinger & Kunzmann, 2005). Roberts and Wood (2006) described the increases in emotional stability, agreeableness, and conscientiousness as the *Maturity Principle* of personality development. Investing in age-graded social roles (*Social Investment Principle*) presumably contributes to personality maturity (Bleidorn, 2015; Roberts, Wood, & Caspi, 2008). During young adulthood investment occurs mainly into new social roles such as spouse, parent, and employee (Hutteman, Hennecke, Orth, Reitz, & Specht, 2014).

In terms of the TESSERA framework (Table 3), new social roles mean that many novel triggering situations and expectancies occur and that these situations and expectancies can elicit personality state intensities outside one's usual range. For example, students transitioning to their first job may be confronted with situations and demands to fulfill demanding tasks reliably, accurately, and in time, and the consequences of (not) doing so are more severe than during school years (Helson, Mitchell, & Moane, 1984). As many of the new social roles during young adulthood result from age-graded life transitions (Havighurst, 1972; Hutteman et al., 2014b), most members of a population display normative patterns of personality development during this period, as described by the maturity principle. At the same time, missing age-graded roles (e.g., spouse, employee) relates to non-normative personality development such as absent increases in emotional stability and conscientiousness (Lehnart et al., 2010; Roberts et al., 2006a). Presumably, trigger situations and expectancies are absent and cannot elicit states, such as being reliable, finishing tasks on time, which would have led to increases in emotional stability and conscientiousness (see Figure 2).

The differential investment into normative experiences and roles contributes to increases in rank-order stability (Lucas & Donellan, 2011, Roberts & DelVecchio, 2000) during young adulthood. Since even rather normative life transitions are partly shaped by traits and retroact on these traits (Hutteman et al., 2014), individual differences in traits consolidate (*Niche-Picking Principle*, Roberts & Damian, in press).

Cumulating Continuity During Middle Adulthood

Middle adulthood is marked by higher rank-order continuity and less mean-level change than other age periods (Lucas & Donnellan, 2011; Roberts & DelVecchio, 2000; Roberts et al., 2006b, van Aken, Denissen, Branje, Dubas, & Goossens, 2006). This pattern is described as the

Cumulative Continuity Principle (Caspi, Roberts, & Shiner, 2005; Roberts & Wood, 2006). The cumulative continuity of personality likely results from largely stable environments and social roles (*Role Continuity Principle*, Roberts & Wood, 2006), increasing self-knowledge (*Identity Development Principle*, Roberts & Wood, 2006), and self-selected life experiences that help to stabilize personality over time (*Niche-Picking Principle*, Roberts & Damian, in press; see also Dickens & Flynn, 2001 for supporting evidence from the cognitive domain). The TESSERA framework helps to detail the short-term processes associated with these principles (Table 3).

Social roles and environments are considerably more stable in middle adulthood compared to earlier times in life because most transitions to new social roles and experiences occur before middle adulthood (Hutteman et al., 2014; Specht et al., 2011). During middle adulthood, social roles (e.g., spouse, parent, or employee) and the accompanying environments (e.g., neighborhood, work place) are maintained for longer periods than in young adulthood (Hutteman et al., 2014b). Consequently, daily lives, the situations found therein, and positive and negative events, are more stable and restricted in variety (Brose, Scheibe, & Schmiedek, 2013; Wrzus, Müller, Wagner, Lindenberger, & Riediger, 2013b; Wrzus et al., 2016). More stable environments and fewer triggering situations that elicit TESSERA sequences could contribute to the greater personality continuity during middle adulthood.

The strengthening of one's identity in middle age could further contribute to greater personality continuity (*Identity Development Principle*, Roberts & Wood, 2006). Throughout the lifespan, people seek, commit to, and develop their identity, their subjective meta-perception of their characteristics. People do so by reflecting on past experiences in light of their current self-view (Swann, 1983; McAdams & Pals, 2006) and by viewing and interpreting new experiences partly in line with existing personality characteristics to achieve a sense of consistency (Caspi & Roberts, 2001; Roberts et al., 2008; Robins & John, 1997). Studies of identity and self-concept show that clarity of one's identity peaks in middle age (Diehl & Hay, 2007; Lodi-Smith & Roberts, 2010). One implication of this increasing psychosocial stability is that TESSERA sequences should be more strongly shaped by current personality characteristics in middle adulthood compared to other periods in life.

In addition to viewing experiences in line with existing personality characteristics, new experiences are also partly selected or evoked in line with one's existing personality characteristics (Buss, 1987; Dickens & Flynn, 2001; Roberts et al., 2008; see also *Niche-Picking Principle* Roberts & Damian, in press). One implication of the *Niche-Picking Principle* is that individuals experience more personality-congruent situations (i.e., triggers) and states with age. Yet so far, this hypothesis is only indirectly supported by research on life experiences (e.g., Denissen, Ulfers, Lüdtke, Muck, & Gerstorf, 2014; Jackson, Thoemmes, Jonkmann, Lüdtke, & Trautwein, 2012; Roberts et al., 2003). An experience-sampling study on personality-situations transactions in daily life confirmed associations between Big Five traits and social situations and activities, but found mixed evidence regarding age differences (Wrzus et al., 2016). Most personality-situation associations did not differ with people's age, some were more pronounced in adolescence and young adulthood and others in late adulthood (Wrzus et al., 2016).

In sum, we suggest that the greater personality continuity in middle adulthood is linked especially to the short-term TESSERA processes of more stable and unchanging situations (Brose et al., 2013), more personality congruent states (Noftle & Fleeson, 2010), and identity-confirming reflections. If more personality-congruent situations are sought in middle adulthood and elicit trait-relevant states, associative processes related to habit formation could further contribute to greater personality

continuity (Rothman et al., 2009; Wood & Neal, 2007).

Plasticity and Diversification During Late Adulthood

During old age, the rank-order stability in personality traits seems to weaken compared to middle adulthood (Ardelt, 2000; Lucas & Donnellan, 2011; Wortman et al., 2012).[9] This likely results from diversification in personality mean-level change (i.e., large variances of mean-level changes, Kandler et al., 2015; Mõttus et al., 2012b; Small, Hertzog, Hultsch, & Dixon, 2003): Many older adults experience declines in personality traits (Gerstorf, Ram, Lindenberger & Smith, 2013; Kandler et al., 2015; Mõttus, Johnson, & Deary, 2012a; Specht et al., 2011; Terracciano, McCrae, Brant, & Costa, 2005), some show no pronounced changes (Mõttus et al., 2012a), and some older adults experience increases in some personality traits (e.g., openness to new experiences; Jackson, Hill, Payne, Roberts, & Stine-Morrow, 2012; Mühlig-Versen, Bowen, & Staudinger, 2012). Thus, the *Plasticity Principle* (Roberts & Wood, 2006), conveying the potential to adapt to changes, also applies to this stage of life, though the most prevalent changes appear to be negative, such as cognitive decline and increasing morbidity and frailty (McArdle, Ferrer-Caja, Hamagami, & Woodcock, 2002; Smith & Baltes, 1997; Steinhagen-Thiessen & Borchelt, 1999). These declines presumably contribute to decreases in personality traits (e.g., Wagner et al., 2015). The TESSERA framework could help to identify the underlying processes of such personality development (Table 3).

Several components of the TESSERA process are likely changed in late adulthood compared to earlier times in life. First, social, health-related, or environmental losses (e.g., death of spouse or friends, illnesses, frailty, moving to a retirement home, giving up activities; Wrzus, Hänel, Wagner, & Neyer, 2013a; Steinhagen-Thiessen & Borchelt, 1999) could entail situations triggering further components of the TESSERA sequence. Diversity in personality development could arise because individuals experience such triggering situations at different times or not at all. Second, expectations about the fulfillment of specific social roles are more open as late adulthood is less structured compared to previous life phases (Freund, Nikitin, & Ritter, 2009). Again, this opens up possibilities for diverse patterns of personality development depending on the social, cognitive, and health-related resources and constraints. Finally, general cognitive decline (Ghisletta, Rabbitt, Lunn, & Lindenberger, 2012; McArdle et al., 2002) also impairs processes of remembering and evaluating events, one's and others' reaction to that event, and learning efficiency (Hofer & Alwin, 2008; Vukman, 2005). Hence, reflective and associative processes could take on unique patterns compared to younger adulthood, again increasing differences in patterns of personality development in late adulthood.

We propose that, in general, young adulthood is the time of greater change, middle adulthood is the time of greater continuity, and late adulthood is the time of diversity and heterogeneity in terms of personality development. The latter has come to focus only recently because personality continuity and change in old age was not researched much as personality was thought to be largely stable after young adulthood (McCrae et al., 2000, but see also McCrae & Costa, 2008). New findings on the diversity of personality trait development beyond young adulthood and the connection of personality with lifespan developmental theories suggest that personality characteristics are subject to the same lifelong plasticity as other domains of human functioning (Baltes, 1987).

Applicability of the TESSERA Framework to Different Personality Domains

Most models of personality incorporate more characteristics than traits such as goals, self-esteem, and life narratives (e.g., Dunlop, 2015; Kandler et al., 2014; McAdams & Olson, 2010; McCrae & Costa Jr, 2008; Roberts & Wood, 2006). We propose a

generic developmental framework applicable to these different personality characteristics, where the specific contents of the TESSERA sequences might differ: For example, relevant features of triggering situations and specific states might differ for the development of constructs like goals or self-esteem (Leary & Baumeister, 2000; Leary & MacDonald, 2003). Also, modifications regarding reflective and associative processes are plausible: Perhaps changes in self-esteem require fewer repetitions of TESSERA sequences because self-esteem is thought to be somewhat more malleable (i.e., less stable) than traits or abilities (Kandler et al., 2014; Leary & Baumeister, 2000; McCrae & Costa Jr, 2008; Roberts & Wood, 2006; Trzesnieski, Donnellan, & Robins, 2003).

A recent longitudinal study on self-esteem change illustrates the applicability of the TESSERA framework for self-esteem development. Increases in state self-esteem during one academic exchange year predicted differential changes in trait self-esteem over the year (Hutteman et al., 2014c). In addition, greater social inclusion prospectively predicted higher state self-esteem and vice versa. From the TESSERA framework's perspective, social inclusion (social activities) triggered state self-esteem, which was reinforced by future social inclusion. This study did not assess reflective processes and thus cannot distinguish between reflective and associative processes as driving forces for trait self-esteem increases.

Also, self-narration, the process for developing life narratives (Hooker & McAdams, 2003b; King, 2001; McLean et al., 2007; Pals, 2006), can be described in terms of the TESSERA framework: Special situations [Trigger] like a disaster, a birthday, or a person asking about important experience can trigger memories of previous experiences [State thought & State expressions of thoughts]. Retelling these memories can garner reactions from other persons (e.g., disdain, humour) or elicit own emotional reactions (e.g., guilt, joy). Pairing these different memories can then lead to a slightly different life story that is told differently the next time. Over time and repeated telling of the story, the life narrative itself changes, partly in content and partly with respect to how it is told (McAdams & Olson, 2010; McLean et al., 2007). As with traits, self-related schemata, and other personality characteristics, the change presumably occurs gradually, not necessarily linearly (see *A Note on the Direction and Time Course of Personality Development Processes*).

An intriguing question concerns the transfer of personality development in one domain to other domains. For example, do increases in self-esteem or in conscientiousness predict subsequent changes in extraversion? First longitudinal evidence finds reciprocal influences among life goals and Big Five traits (Bleidorn et al., 2010) and changes in self-esteem linked to changes in Big Five traits (Wagner, Lüdtke, Jonkmann, & Trautwein, 2013). In contrast, changes in Big Five traits themselves seem largely uncorrelated (Klimstra, Bleidorn, Asendorpf, van Aken, & Denissen, 2013; Mõttus et al., 2012b; Soto & John, 2012). Both patterns are consistent with the TESSERA framework. Correlated change among personality characteristics could occur if situations elicit states associated with different characteristics. For example, if repeated pleasant social interactions elicit both higher state self-esteem and higher extraverted behaviour repeatedly, but not orderly behaviour, both self-esteem and extraversion, but not conscientiousness could increase. This view is consistent with functional perspectives (Wood et al., 2015) and network approaches to personality (Cramer et al., 2012): Certain states are more likely to co-occur because they are linked to the same eliciting situation or fulfil compatible functions, whereas other states often preclude each other (e.g., being talkative and getting things done efficiently). Hence, we propose that correlated change in different personality domains can occur if the corresponding states are elicited and reinforced concurrently and repeatedly.

In sum, we propose that the TESSERA framework is equally valuable for studying and understanding the development of other

personality characteristics such as self-esteem, goals, or life narratives. In describing the framework and its moderators, we focused on behavioral dispositions because of the richer empirical evidence for continuity and change in this domain. Similar developmental research on the longitudinal development of goals or life-narratives is currently still scarce (e.g., Bleidorn et al., 2010, Lüdtke et al., 2009; Roberts, O'Donnell, & Robins, 2004), especially regarding people in middle and late adulthood (Wrzus & Lang, 2010). Furthermore, as traits, motivational orientations, and life narrative all reside within the same person, examining their reciprocal influence would advance the holistic understanding of personality development (Dunlop, 2015). The TESSERA framework offers conceptualizations on the short- and long-term processes underlying such longitudinal, concurrent development.

Contribution of the TESSERA Framework and Unique Predictions

We see four main contributions of the TESSERA framework for the study of personality development. First, the framework advances previous work, such as the *Social Investment Principle* (Lodi-Smith & Roberts, 2007) and the *Neo-Socioanalytic Model of Personality* (Roberts & Wood, 2006) by focusing on short-term processes and their linkage to long-term development. The TESSERA framework resorts to social-cognitive-behavioral models of functioning, yet, such models did not focus on personality development (e.g., Ajzen, 1991; Back et al., 2011; Cervone, 2005; Fleeson, 2012; Kanfer & Phillips, 1970; Mischel & Shoda, 1995, 1998). For example, the CAPS model (Mischel & Shoda, 1995, 1998), which aims at explaining personality and behavioral consistencies, also includes situations and expectancies. Similarly, the PERSOC framework (Back et al., 2011, which focuses on how personality affects interpersonal perception and social behavior, views other people's behavior as situational cues for interpersonal perception and behavior, which can be reinforced by others. Both models briefly address the development of personality traits through repeated experiences (Back et al., 2009, 2011; Mischel & Shoda, 1995). However, neither of these models discusses the role of antecedents, consequences, and moderators for personality development or distinguish associative and reflective processes. Thus, the TESSERA merges a social-cognitive model of personality with a trait model of personality development, which while argued for in many circles (e.g., Roberts, 2009), has not yet been fully accomplished.

Second, the TESSERA identifies specific components of the process that are not addressed and organized in prior theories of personality development (e.g., Hennecke et al., 2014; Roberts & Wood, 2006). Accordingly, measurement burst study designs (Nesselroade, 2004; Sliwinski, 2008) that assess personality characteristics longitudinally and multi-methodologically as well as all short-term components, associative and reflective processes repeatedly would be desirable if not necessary to test the TESSERA framework. For example, Wrzus et al. (2016) linked repeated sequences of momentary hassle situations (triggers) eliciting higher negative affect (state) to long-term change in emotional stability over six years. The mere repeated occurrence of hassles did not predict changes in emotional stability. Hutteman et al (2014c) linked repeated monthly experiences of social inclusion and state self-esteem to change in trait self-esteem over one year. Yet, adding assessments of expectancies, own and others' reactions in such situations, different types of reflection about the experiences, as well as implicit or behavioral manifestations of personality characteristics would offer a better understanding on the long-term change in both studies. That means, the current framework specifies exact groups of variables that should be assessed, but researchers nonetheless adjust the variables to the specific personality characteristic being studied (e.g., traits, life narratives).

Third, the generic nature of the TESSERA framework leads to the identification of

common processes and potential moderators that previous theorizing could not address due to focusing only on specific domains of personality development (e.g., narrative development McLean et al., 2007; motivated personality trait development, Hennecke et al., 2014; Wood & Denissen, 2015). Through application to various personality characteristics (e.g., traits, goals, life narratives), developmental processes in these characteristics can be compared and linked. For example, considering the TESSERA processes invites questions about when developmental change needs to manifest through explicit factors, implicit factors, or both. For example, are some characteristics, such as self-esteem, more strongly changed through reflective compared to associative processes? Do personality traits and goals differ in the number of repeated TESSERA sequences needed for eliciting personality development? To what extent do the moderators, such as external-internal locus, affect development of certain characteristics differently? Comparing the developmental processes of different personality characteristics by using one framework could provide a more comprehensive understanding the general and unique conditions under which different aspects of personality develop (Hooker & McAdams, 2003a; Hooker & McAdams, 2003b).

Finally, the TESSERA framework leads to novel insights into existing developmental phenomena, which were not addressed in previous theories (e.g., Back et al., 2011a; Hennecke et al., 2014; Mischel & Shoda, 1995; Roberts & Wood, 2006), such as why correlated change among Big Five traits and motivational orientations differ and why different patterns of personality continuity and change are observed across the adult lifespan. In addition to normative patterns of personality development due to normative life transitions with similar situations and expectancies (Bleidorn et al., 2013; Neyer et al., 2014), individual differences in the strength and direction of development can be explained through differences in personal or environmental factors that affect the TESSERA components.

We conclude by presenting three testable sets of predictions that are derived uniquely from the TESSERA framework:

- Repeated TESSERA sequences lead to personality development. Reflective processes that reduce the discrepancy between the momentary expectancies or states and one's current personality characteristics should predict changes in explicit representations and these changes should be stronger when compared to implicit representations. In contrast, associative processes (i.e., the number of state repetitions together with reactions) should predict changes in implicit representations and these changes should be stronger when compared to explicit representations. Previous theories predicted that repeated states predict personality trait development (Back et al., 2011; Mischel & Shoda, 1995; Roberts, 2006, 2009), hence did not specify associative and reflective processes and did not distinguish development of implicit, behavioral, and explicit trait manifestations.
- Substantial personality change can occur in middle and late adulthood, if (a) enough "opportunities" to experience new situations exist (e.g., life experiences typically experienced during young adulthood, entering the work force), (b) more repetitions occur to elicit changes in implicit associations compared to young adulthood, and (c) novel states are reflected as novel instead of personality-congruent like during identity confirmation. At the same time, new situations might affect older people less because older people's generally higher level of maturity (emotional stability, conscientiousness) facilitates better adaption to the demands of the change.

  Previous theories on personality development attributed the less pronounced personality change in

middle and late adulthood to a more consolidated identity and an increased personality-environment fit (Caspi & Roberts, 1999, 2001; Roberts & Wood, 2006), but the theories made no predictions when substantial change could occur.

- Correlated change will be stronger among personality domains that share similar situations and states compared to conflicting situations and states. For example, correlated changes in affiliative goals, extraversion, and general self-esteem should be stronger compared to correlated changes in work-related goals and general self-esteem or conscientiousness and self-esteem because the former relate to social situations and social behavior. Previous theories focused on correlated change among (Big Five) traits and did not consider other aspects of personality, such as motivational orientations. The studies either attributed correlated change to a shared underlying life transition or biological mechanism (Klimstra et al., 2013), or to a shared functionality (i.e., Wood et al., 2015).

Future Research Directions

Despite the comprehensive scope of linking short- and long-term changes in personality characteristics, addressing associative as well as reflective processes, and including behavioral, emotional, cognitive, and motivational domains, a number of limitations of the TESSERA framework should be considered. First, biological factors of personality development are only implicitly included in the framework. We discussed characteristics of individuals, such as traits, goals, cognitive abilities, and health, as potential moderators of the TESSERA processes. Undisputable these characteristics have neurobiological and hence genetic underpinnings (e.g., Bouchard & McGue, 1981; DeYoung & Gray, 2009; Johnson, Vernon, & Feiler, 2008; Tucker-Drob, Briley, & Harden, 2013). Accordingly, neurobiological and genetic differences can be viewed as moderators of the TESSERA processes during personality development, albeit on a different measurement and functional level than psychological constructs. Recent studies showed that the genetic influences on personality continuity (i.e., rank-order stability) and differences in personality mean-level changes are important but likely decrease over the lifespan (Bleidorn, Kandler, & Caspi, 2014; Briley & Tucker-Drob, 2015; Kandler et al., 2015). Environmental influences become more important for both personality continuity and differences in mean-level changes (Bleidorn et al., 2014; Bleidorn, Kandler, Riemann, Angleitner, & Spinath, 2009; Blonigen, Littlefield, Hicks, & Sher, 2010; Briley & Tucker-Drob, 2015; Kandler et al., 2010; Loehlin, 1992). Still, the relative importance of genetic and environmental factors over the lifespan for personality development is difficult to judge because genetic factors contribute to environmental influences (e.g., life transitions, Bemmels, Burt, Legrand, Iacono, & McGue, 2008) and environmental factors affect the influence of genetic factors (e.g., gene-environment transactions, Caspi et al., 2005, Dickens & Flynn, 2001; Roberts & Jackson, 2008; Tucker-Drob & Briley, 2014). Further work is needed that details processes how biophysiological factors (including genetic variation) contribute to psychological processes of personality development, but also how psychological processes affect biophysiological factors (Roberts & Jackson, 2008; Slavich & Irwin, 2014).

Second, changes in personality characteristics perceived by other people are not depicted in the TESSERA framework. Personality characteristics can be validly assessed using informant reports (e.g., reports from romantic partners, friends, or family members; Soto & John, 2009; John et al., 2008). Developmental changes based on informant reports largely mirror the patterns derived from self-reports of personality characteristics (McCrae et al., 2004), and presumably result from repeatedly observing

the individuals' behavior in relevant triggering situations (Back et al., 2011a). Thus, the informant (e.g., romantic partner, friend) observes behavioral change and integrates these perceptions into his/her view of the individual (i.e., reputation, Back et al., 2011a). The interpersonal perceptions and their integration into the reputation can also be subject to perception biases (e.g., Leising, Gallrein, & Dufner, 2014; Letzring, 2008; Wood, Brown, Maltby, & Watkinson, 2012). Accordingly, the TESSERA framework could be expanded to changes in reputation, in addition to explicit and implicit representations. Yet, the expended framework would need to consider the personality of the informant as well as perceiver, target and relationship effects of the interpersonal perception and judgment processes (see PERSOC framework, Back et al., 2011a).

Third, we emphasized that long-term personality development should occur as a result of short-term, situational processes lasting only a few minutes and proposed TESSERA sequences as such processes. However, these short-term processes likely have counterparts on the microprocess level (Nesselroade & Molenaar, 2010). In the study on reducing aggression by viewing and interpreting ambiguous faces (Penton-Voak et al., 2013) state anger could be scrutinized regarding microprocesses in face perception and anger elicitation. Similarly, recent theoretical work linked stress reactions to depressive behavior through inflammatory processes on the molecular level (Slavich & Irwin, 2014). Future research might target how the encoding of psychologically relevant situational features, the elicitation of behavior, or affective reactions happen on the micro-level of perceptual or biological processes (e.g., in milliseconds). Linking such micro-level processes to long-term development seems a very challenging endeavor at the moment, since we are still struggling with understanding the short-term processes.

Fourth, future research also could look beyond individual development and address how societal factors influence the TESSERA framework of personality development. We hypothesize that the general TESSERA process is comparable across cultures and cohorts, but the components may differ in level and emphasis. For example, expectations on how to behave, feel, and express feelings differs across cultures and for instance varies with dimensions of individualism—collectivism (Oyserman, Coon, & Kemmelmeier, 2002; Ward et al., 2004).

In sum, we have proposed a framework on personality development that attributes long-term personality development to repeated short-term, so-called TESSERA sequences. According to the proposed framework sequences of Triggering situations, Expectancies, States/State expressions, and Reactions lead up to personality development through associative and reflective processes concerning TESSERA sequences. Review of previous theoretical and empirical work on personality development led to the identification of the components of this generic framework. The framework is capable of explaining both normative patterns and individual differences in personality development on different levels: traits, motivational orientations, and life narratives. Now it awaits empirical work to test its assumptions and provide data on which to evaluate the model, which hopefully lead to positive developments in the coming years.

References

Aarts, H., & Dijksterhuis, A. (2000). Habits as knowledge structures: Automaticity in goal-directed behavior. *Journal of Personality and Social Psychology, 78*, 53-63. doi: 10.1037/0022-3514.78.1.53

Adler, N. E., & Newman, K. (2002). Socioeconomic disparities in health: Pathways and policies. *Health affairs, 21*, 60-76.

Asendorpf, J. B., Banse, R., & Mücke, D. (2002). Double dissociation between implicit and explicit personality self-concept: the case of shy behavior. *Journal of Personality and Social Psychology, 83*, 380-393. doi: 10.1037//0022-3514.83.2.380

Ajzen, I. (1991). The theory of planned behavior. *Organizational behavior and human decision processes, 50*, 179-211.

Ardelt, M. (2000). Still stable after all these years? Personality stability theory revisited. *Social Psychology Quarterly, 63*, 392-405.

Back, M. D., Baumert, A., Denissen, J. J. A., Hartung, F.-M., Penke, L., Schmukle, S. C., Schönbrodt, F. D., Schröder-Abé, M., Vollmann, M., Wagner, J., & Wrzus, C. (2011a). PERSOC: A unified framework for understanding the dynamic interplay of personality and social relationships. *European Journal of Personality, 25*, 90-107. doi: 10.1002/per.811

Back, M. D., & Nestler, S. (in press). Dual process approaches to personality. In R. Deutsch, B. Gawronski & W. Hofmann (Eds.), *Reflective and impulsive determinants of human behavior* (pp. nn). New York: Psychology Press.

Back, M. D., Penke, L., Schmukle, S. C., Sachse, K., Borkenau, P., & Asendorpf, J. B. (2011b). Why mate choices are not as reciprocal as we assume: The role of personality, flirting and physical attractiveness. *European Journal of Personality, 25*, 120-132.

Back, M. D., Schmukle, S. C., & Egloff, B. (2009). Predicting actual behavior from the explicit and implicit self-concept of personality. *Journal of Personality and Social Psychology, 97*, 533-548. doi: 10.1037/a0016229

Back, M. D., & Vazire, S. (2012). Knowing our personality. In S. Vazire & T. D. Wilson (Eds.), *Handbook of self-knowledge* (pp. 131 – 156). New York: Guilford.

Baltes, P. B. (1987). Theoretical propositions of life-span developmental psychology: On the dynamics between growth and decline. *Developmental Psychology, 23*, 611-626.

Bandura, A. (1977). Self-efficacy: Toward a unifying theory of behavioral change. *Psychological Review, 84*, 191-215.

Bandura, A. (1986). *Social foundations of thought and action*. Englewood Cliffs, NJ: Prentice-Hall.

Bandura, A. (2003). Observational learning. In J. H. Byrne (Ed.), *Encyclopedia of learning and memory* (2nd ed., pp. 482-484). New York: Macmillan.

Bargh, J. A., & Ferguson, M. J. (2000). Beyond behaviorism: On the automaticity of higher mental processes. *Psychological Bulletin, 126*, 925-945.

Baumeister, R. F., Bratslavsky, E., Finkenauer, C., & Vohs, K. D. (2001). Bad is stronger than good. *Review of General Psychology, 5*, 323-370. doi: 10.1037//1089-2680.5.4.323

Bem, D. J. (1972). *Self-perception theory*. New York: Academic Press.

Bemmels, H. R., Burt, S. A., Legrand, L. N., Iacono, W. G., & McGue, M. (2008). The heritability of life events: An adolescent twin and adoption study. *Twin Research and Human Genetics, 11*, 257-265. doi: doi:10.1375/twin.11.3.257

Berg, A. I., & Johansson, B. (2014). Personality change in the oldest-old: Is it a matter of compromised health and functioning? *Journal of Personality, 82*, 25-31. doi: 10.1111/jopy.12030

Bisconti, T. L., Bergeman, C., & Boker, S. M. (2004). Emotional well-being in recently bereaved widows: A dynamical systems approach. *The Journals of Gerontology, 59B*, 158-167. doi: 10.1093/geronb/59.4.P158

Bleidorn, W. (2012). Hitting the road to adulthood: Short-term personality development during a major life transition. *Personality and Social Psychology Bulletin, 38*, 1594-1608. doi: 10.1177/0146167212456707

Bleidorn, W. (2015). What accounts for personality maturation in early adulthood?

Current Directions in Psychological Science, 24, 245-252. doi: 10.1177/0963721414568662

Bleidorn, W., Kandler, C., & Caspi, A. (2014). The behavioural genetics of personality development in adulthood—Classic, contemporary, and future trends. *European Journal of Personality, 28*, 244-255

Bleidorn, W., Kandler, C., Hülsheger, U. R., Riemann, R., Angleitner, A., & Spinath, F. M. (2010). Nature and nurture of the interplay between personality traits and major life goals. *Journal of Personality and Social Psychology, 99*, 366-379. doi: 10.1037/a0019982

Bleidorn, W., Kandler, C., Riemann, R., Angleitner, A., & Spinath, F. M. (2009). Patterns and sources of adult personality development: Growth curve analyses of the NEO PI-R scales in a longitudinal twin study. *Journal of Personality and Social Psychology, 97*, 142-155. doi: 10.1037/a0015434

Bleidorn, W., Klimstra, T. A., Denissen, J. J. A., Rentfrow, P. J., Potter, J., & Gosling, S. D. (2013). Personality maturation around the world: A cross-cultural examination of social-investment theory. *Psychological Science, 24*, 2530-2540. doi: 10.1177/0956797613498396

Block, J. (1982). Assimilation, accommodation, and the dynamics of personality development. *Child Development*, 281-295.

Blonigen, D. M., Littlefield, A. K., Hicks, B. M., & Sher, K. J. (2010). Course of antisocial behavior during emerging adulthood: Developmental differences in personality. *Journal of Research in Personality, 44*, 729-733. doi: 10.1016/j.jrp.2010.08.008

Bouchard, T. J., & McGue, M. (1981). Familial studies of intelligence: A review. *Science, 212*(4498), 1055-1059

Brandtstädter, J. (1989). Personal self-regulation of development: Cross-sequential analyses of development-related control beliefs and emotions. *Developmental Psychology, 25*, 96-108.

Brandtstädter, J., & Greve, W. (1994). The aging self: Stabilizing and protective processes. *Developmental Review, 14*, 52-80

Brandtstädter, J., & Renner, G. (1990). Tenacious goal pursuit and flexible goal adjustment: Explication and age-related analysis of assimilative and accommodative strategies of coping. *Psychology and Aging, 5*, 58-67.

Briley, D. A., & Tucker-Drob, E. M. (2015). Comparing the developmental genetics of cognition and personality over the life span. *Journal of Personality*, nn. doi: 10.1111/jopy.12186

Briñol, P., Petty, R. E., & Wheeler, S. C. (2006). Discrepancies between explicit and implicit self-concepts: Consequences for information processing. *Journal of Personality and Social Psychology, 91*, 154-170. doi: 10.1037/0022-3514.91.1.154

Brose, A., Scheibe, S., & Schmiedek, F. (2013). Life contexts make a difference: Emotional stability in younger and older adults. *Psychology and Aging, 28*, 148-159. doi: 10.1037/a0030047

Burke, P. J. (1991). Identity processes and social stress. *American Sociological Review, 56*, 836-849.

Buss, D. M. (1987). Selection, evocation, and manipulation. *Journal of Personality and Social Psychology, 53*, 1214-1221.

Buss, D. M., & Craik, K. H. (1983). The act frequency approach to personality. *Psychological Review, 90*, 105-126. doi: 10.1037/0033-295X.90.2.105

Carver, C. S. (2004). Negative affect deriving from the behavioral approach system. *Emotion, 4*, 3-22. doi: 10.1037/1528-3542.4.1.3

Caspi, A., & Roberts, B. W. (1999). Personality continuity and change across the life course. In L. A. Pervin, & O. P. John (Eds.), *Handbook of personality: Theory and research (2nd ed.).* (pp. 300-326): Guilford Press.

Caspi, A., & Roberts, B. W. (2001). Personality development across the life course: The argument for change and continuity. *Psychological Inquiry, 12*, 49-66.

Caspi, A., Roberts, B. W., & Shiner, R. L. (2005). Personality development: Stability and change. *Annual Review of Psychology, 56*, 453-484.

Cervone, D. (2005). Personality architecture: Within-person structures and processes. *Annual Review of Psychology, 56*, 423-452.

Chow, P. I., & Roberts, B. W. (2014). Examining the relationship between changes in

personality and changes in depression. *Journal of Research in Personality, 51,* 38-46. doi: 10.1016/j.jrp.2014.04.007

Cramer, A. O., Sluis, S., Noordhof, A., Wichers, M., Geschwind, N., Aggen, S. H., ... & Borsboom, D. (2012). Dimensions of normal personality as networks in search of equilibrium: You can't like parties if you don't like people. *European Journal of Personality, 26,* 414-431

Datan, N., Rodeheaver, D., & Hughes, F. (1987). Adult development and aging. *Annual Review of Psychology, 38,* 153-180.

De Fruyt, F., Van Leeuwen, K., Bagby, R. M., Rolland, J. P., & Rouillon, F. (2006). Assessing and interpreting personality change and continuity in patients treated for major depression. *Psychological Assessment, 18,* 71-80. doi: 10.1037/1040-3590.18.1.71

De Fruyt, J., Deschepper, E., Audenaert, K., Constant, E., Floris, M., Pitchot, W., ... & Claes, S. (2012). Second generation antipsychotics in the treatment of bipolar depression: a systematic review and meta-analysis. *Journal of Psychopharmacology, 26,* 603-617. 0269881111408461.

Denissen, J. J. A., van Aken, M. A. G., Penke, L., & Wood, D. (2013). Self-regulation underlies temperament and personality: An integrative developmental framework. *Child Development Perspectives, 7,* 55-60. doi: 10.1111/cdep.12050

Denissen, J. J. A., Ulfers, H., Lüdtke, O., Muck, P. M., & Gerstorf, D. (2014). Longitudinal transactions between personality and occupational roles: A large and heterogeneous study of job beginners, stayers, and changers. *Developmental Psychology, 50,* 1931-1942. doi: 10.1037/a0036994

Deutsch, R., & Strack, F. (2006). Duality models in social psychology: From dual processes to interacting systems. *Psychological Inquiry, 17,* 166-172.

DeYoung, C. G. (2015). Cybernetic Big Five Theory. *Journal of Research in Personality, 56,* 33-58. doi: 10.1016/j.jrp.2014.07.004

DeYoung, C. G., & Gray, J. R. (2009). Personality neuroscience: Explaining individual differences in affect, behavior, and cognition. In P. J. Corr & G. Matthews (Eds.), *The Cambridge handbook of personality psychology* (pp. 323–346). New York: Cambridge University Press.

Dickens, W. T., & Flynn, J. R. (2001). Heritability estimates versus large environmental effects: The IQ paradox resolved. *Psychological Review, 108,* 346-369. doi: 10.1037//0033-295X. 108.2.346

Diehl, M., & Hay, E. L. (2007). Contextualized self-representations in adulthood. *Journal of Personality, 75,* 1255-1284.

Dunlop, W. L. (2015). Contextualized personality, beyond traits. *European Journal of Personality, 29,* 310-325. doi: 10.1002/per.1995

Durbin C. E., & Hicks B. M. (2014). Personality and psychopathology: A stagnant field in need of development, *European Journal of Personality, 28,* 362–386. doi: 10.1002/per.1962

Durbin, C. E., Hicks, B. M., Blonigen, D., & Johnson, W. (2016). Personality trait change across late childhood to young adulthood: Evidence for nonlinearity and sex differences in change. *European Journal of Personality, 30,* 31-44. doi: 10.1002/per.2013

Egloff, B., Weck, F., & Schmukle, S. C. (2008). Thinking about anxiety moderates the relationship between implicit and explicit anxiety measures. *Journal of Research in Personality, 42,* 771-778. doi: 10.1016/j.jrp.2007.08.005

Filipp, S.-H. (1990). Ein allgemeines Modell für die Analyse kritischer Lebensereignisse [A general model for the analysis of critical life events]. In S.-H. Filipp (Ed.), *Kritische Lebensereignisse [Critical life events]* (2., erw. Aufl. ed., pp. 3-52). München Psychologie Verlags Union.

Finn, C., Mitte, K., & Neyer, F. J. (2015). Recent decreases in specific interpretation biases predict decreases in neuroticism: Evidence from a longitudinal study with young adult couples. *Journal of Personality, 83,* 274–286. doi: 10.1111/jopy.12102

Flavell, J. H. (2000). Development of children's knowledge about the mental world. *International Journal of Behavioral Development, 24,* 15-23.

Fleeson, W. (2001). Toward a structure- and process-integrated view of personality: Traits as density distributions of states. *Journal of*

Personality and Social Psychology, 80, 1011-1027. doi: 10.1037//0022-3514.80.6.1011

Fleeson, W. (2012). Perspectives on the person: Rapid growth and opportunities for integration. In K. Deaux & M. Snyder (Eds.), *The Oxford handbook of personality and social psychology*. (pp. 33-63). Oxford: Oxford University Press.

Fleeson, W., & Gallagher, P. (2009). The implications of Big Five standing for the distribution of trait manifestation in behavior: Fifteen experience-sampling studies and a meta-analysis. *Journal of Personality and Social Psychology, 97*, 1097-1114.

Fleeson, W., & Jayawickreme, E. (2015). Whole trait theory. *Journal of Research in Personality*. doi: 10.1016/j.jrp.2014.10.009

Fleeson, W., & Jolley, S. (2006). A proposed theory of the adult development of intraindividual variability in trait-manifesting behavior. In D. Mroczek, & T. D. Little (Eds.), *Handbook of Personality Development* (pp. 41-59). Mahwah, NJ: Lawrence Erlbaum Associates.

Flückiger, C., Grosse Holtforth, M., Del Re, A., & Lutz, W. (2013). Working along sudden gains: Responsiveness on small and subtle early changes and exceptions. *Psychotherapy, 50*, 292-297. doi: 10.1037/a0031940

Freund, A. M., Nikitin, J., & Ritter, J. O. (2009). Psychological consequences of longevity: The increasing importance of self-regulation in old age. *Human Development, 52*, 1-37.

Gawronski, B., & Bodenhausen, G. V. (2006). Associative and propositional processes in evaluation: an integrative review of implicit and explicit attitude change. *Psychological Bulletin, 132*, 692-731. doi: 10.1037/0033-2909.132.5.692

Gerlsma, C., Emmelkamp, P. M., & Arrindell, W. A. (1990). Anxiety, depression, and perception of early parenting: A meta-analysis. *Clinical Psychology Review, 10*, 251-277.

Gerstorf, D., Hoppmann, C. A. & Ram, N. (2014) The promise and challenges of integrating multiple time-scales in adult developmental inquiry. *Research in Human Development, 11*, 75-90, doi: 10.1080/15427609.2014.906725

Gerstorf, D., Ram, N., Goebel, J., Schupp, J., Lindenberger, U., & Wagner, G. G. (2010). Where people live and die makes a difference: Individual and geographic disparities in well-being progression at the end of life. *Psychology and Aging, 25*, 661.

Gerstorf, D., Ram, N., Lindenberger, U., & Smith, J. (2013). Age and time-to-death trajectories of change in indicators of cognitive, sensory, physical, health, social, and self-related functions. *Developmental Psychology, 49*, 1805-1821. doi: 10.1037/a0031340

Ghisletta, P., Rabbitt, P. M. A., Lunn, M., & Lindenberger, U. (2012). Two thirds of the age-based changes in fluid and crystallized intelligence, perceptual speed, and memory in adulthood are shared. *Intelligence, 40*, 260-268.

Greve, W., Rothermund, K., & Wentura, D. (2005). *The adaptive self: Personal continuity and intentional self-development*. Göttingen: Hogrefe.

Hallion, L. S., & Ruscio, A. M. (2011). A meta-analysis of the effect of cognitive bias modification on anxiety and depression. *Psychological Bulletin, 137*, 940-958. doi: 10.1037/a0024355

Havighurst, R. J. (1972). *Developmental tasks and education*. New York: McKay Company

Heckhausen, H., & Gollwitzer, P. M. (1987). Thought contents and cognitive functioning in motivational versus volitional states of mind. *Motivation and Emotion, 11*, 101–120. doi: 10.1007/BF00992338

Heckhausen, J., Wrosch, C., & Schulz, R. (2010). A motivational theory of life-span development. *Psychological Review, 111*, 32-60.

Helson, R., Mitchell, V., & Moane, G. (1984). Personality and patterns of adherence and nonadherence to the social clock. *Journal of Personality and Social Psychology, 46*, 1079-1096. doi: 10.1037/0022-3514.46.5.1079

Hennecke, M., Bleidorn, W., Denissen, J. J., & Wood, D. (2014). A three-part framework for self-regulated personality development across adulthood. *European Journal of Personality, 28*, 289-299

Higgins, E. T. (1996). Knowledge activation: Accessibility, applicability, and salience. In E. T. Higgins & A. W. Kruglanski (Eds.), *Social*

psychology: Handbook of basic principles (pp. 133-168). New York: Guilford Press.

Hofer, S. M., & Alwin, D. F. (Eds.). (2008). *Handbook of cognitive aging: Interdisciplinary perspectives*. California, USA: Sage Publications.

Hofmann, W., Gawronski, B., Gschwendner, T., Le, H., & Schmitt, M. (2005). A meta-analysis on the correlation between the implicit association test and explicit self-report measures. *Personality and Social Psychology Bulletin, 31*, 1369-1385.

Holmes, T. H., & Rahe, R. H. (1967). The social readjustment rating scale. *Journal of Psychosomatic Research, 11*, 213-218. doi: 10.1016/0022-3999(67)90010-4

Hooker, K., & McAdams, D. P. (2003a). Personality and adult development: Looking beyond the OCEAN. *Journals of Gerontology, 58B*, 311-312.

Hooker, K., & McAdams, D. P. (2003b). Personality reconsidered: A new agenda for aging research. *Journals of Gerontology, 58B*, 296-304.

Hopwood, C. J., Newman, D. A., Donnellan, M. B., Markowitz, J. C., Grilo, C. M., Sanislow, C. A., ... & Morey, L. C. (2009). The stability of personality traits in individuals with borderline personality disorder. *Journal of abnormal psychology, 118*, 806-815. doi:10.1037/a0016954

Hudson, N. W., & Fraley, R. C. (2015). Volitional personality trait change: Can people choose to change their personality traits? *Journal of Personality and Social Psychology*. doi: 10.1037/pspp0000021

Hudson, N. W., & Roberts, B. W. (2014). Goals to change personality traits: Concurrent links between personality traits, daily behavior, and goals to change oneself. *Journal of Research in Personality, 53*, 68-83. doi: 10.1016/j.jrp.2014.08.008

Hudson, N. W., Roberts, B. W., & Lodi-Smith, J. (2012). Personality trait development and social investment in work. *Journal of Research in Personality, 46*, 334-344.

Hutteman, R., Bleidorn, W., Keresteš, G., Brković, I., Butković, A., & Denissen, J. J. A. (2014a). Reciprocal associations between parenting challenges and parents' personality development in young and middle adulthood. *European Journal of Personality, 28*, 168-179. doi: 10.1002/per.1932

Hutteman, R., Hennecke, M., Orth, U., Reitz, A. K., & Specht, J. (2014b). Developmental tasks as a framework to study personality development in adulthood and old age. *European Journal of Personality, 28*, 267-278

Hutteman, R., Nestler, S., Wagner, J., Egloff, B., & Back, M. D. (2014c). Wherever I may roam: Processes of self-esteem development from adolescence to emerging adulthood in the context of international student exchange. *Journal of Personality and Social Psychology. 108*, 767-783. doi: 10.1037/pspp0000015

Jackson, J. J., Hill, P. L., Payne, B. R., Roberts, B. W., & Stine-Morrow, E. A. L. (2012). Can an old dog learn (and want to experience) new tricks? Cognitive training increases openness to experience in older adults. *Psychology and Aging, 27*, 286-292. doi: 10.1037/a0025918

Jackson, J. J., Thoemmes, F., Jonkmann, K., Lüdtke, O., & Trautwein, U. (2012). Military training and personality trait development: Does the military make the man, or does the man make the military? *Psychological Science, 23*, 270-277. doi: 10.1177/0956797611423545

Jayawickreme, E., & Blackie, L. E. (2014). Post-traumatic growth as positive personality change: Evidence, controversies and future directions. *European Journal of Personality, 28*, 312-331.

John, O. P., Naumann, L. P., & Soto, C. J. (2008). Paradigm shift to the integrative Big Five trait taxonomy: History, measurement, and conceptual issues. In O. P. John, R. W. Robins & L. A. Pervin (Eds.), *Handbook of personality: Theory and research* (Vol. 3, pp. 114-158). New York, NY: Guilford Press.

Johnson, A. M., Vernon, P. A., & Feiler, A. R. (2008). Behavioral genetic studies of personality: An introduction and review of the results of 50+ years of research. In G. J. Boyle, G. Matthews, & D. H. Saklofske (Eds.), *The SAGE handbook of personality theory and assessment: Volume 1 personality theories and models* (pp. 145–173). London: SAGE.

Jongedijk, R., Carlier, I. E., Schreuder, B. N., & Gersons, B. R. (1996). Complex posttraumatic Stress disorder: An exploratory investigation

of PTSD and DES NOS among Dutch war veterans. *Journal of Traumatic Stress*, 577-586. doi: 10.1007/bf02103666

Kandler, C., Bleidorn, W., Riemann, R., Spinath, F. M., Thiel, W., & Angleitner, A. (2010). Sources of cumulative continuity in personality: A longitudinal multiple-rater twin study. *Journal of Personality and Social Psychology, 98*, 995-1008. doi: 10.1037/a0019558

Kandler, C., Kornadt, A. E., Hagemeyer, B., & Neyer, F. J. (2015). Patterns and sources of personality development in old age. *Journal of Personality and Social Psychology, 109*, 175-191. doi: 10.1037/pspp0000028

Kandler, C., Zimmermann, J., & McAdams, D. P. (2014). Core and surface characteristics for the description and theory of personality differences and development. *European Journal of Personality, 28*, 231-243. doi: 10.1002/per.1952

Kanfer, F. H., & Grimm, L. G. (1977). Behavioral analysis selecting target behaviors in the interview. *Behavior Modification, 1*, 7-28

Kanfer, F. H., & Phillips, J. S. (1970). *Learning foundations of behavior therapy*. New York: Wiley.

King, L. A. (2001). The hard road to the good life: The happy, mature person. *Journal of Humanistic Psychology, 41*, 51-72.

Klimstra, T. A., Bleidorn, W., Asendorpf, J. B., van Aken, M. A. G., & Denissen, J. J. A. (2013). Correlated change of Big Five personality traits across the lifespan: A search for determinants. *Journal of Research in Personality, 47*, 768-777. doi: 10.1016/j.jrp.2013.08.004

Krampen, G. (1988). Toward an action-theoretical model of personality. *European Journal of Personality, 2*, 39-55. doi: 10.1002/per.2410020104

Kwan, V. S., John, O. P., Kenny, D. A., Bond, M. H., & Robins, R. W. (2004). Reconceptualizing individual differences in self-enhancement bias: An interpersonal approach. *Psychological Review, 111*, 94-110. doi: 10.1037/0033-295X.111.1.94

Le, K., Donnellan, M. B., & Conger, R. (2014). Personality development at work: Workplace conditions, personality changes, and the corresponsive principle. *Journal of Personality, 82*, 44-56.

Leary, M. R., & Baumeister, R. F. (2000). The nature and function of self-esteem: Sociometer theory. In M. P. Zanna (Ed.), *Advances in experimental social psychology*: Vol. 32 (pp. 1–62). San Diego: Academic Press.

Leary, M. R., & MacDonald, G. (2003). Individual differences in self-esteem: A review and theoretical integration. In M. R. Leary & J. P. Tangney (Eds.), *Handbook of self and identity* (pp. 401–418). New York: Guilford Press.

Lehnart, J., Neyer, F. J., & Eccles, J. (2010). Long-term effects of social investment: The case of partnering in young adulthood. *Journal of Personality, 78*, 639-670.

Leising, D., Gallrein, A.-M. B., & Dufner, M. (2014). Judging the behavior of people we know: Objective assessment, confirmation of preexisting views, or both? *Personality and Social Psychology Bulletin, 40*, 153-163. doi: 10.1177/0146167213507287

Letzring, T. D. (2008). The good judge of personality: Characteristics, behaviors, and observer accuracy. *Journal of Research in Personality, 42*, 914-932.

Lickel, B., Kushlev, K., Savalei, V., Matta, S., & Schmader, T. (2014). Shame and the motivation to change the self. *Emotion, 14*, 1049-1061. doi: 10.1037/a0038235

Lodi-Smith, J., & Roberts, B. W. (2007). Social investment and personality: A meta-analysis of the relationship of personality traits to investment in work, family, religion, and volunteerism. *Personality and Social Psychology Review, 11*, 1-19.

Lodi-Smith, J., & Roberts, B. W. (2010). Getting to know me: Social role experiences and age differences in self-concept clarity during adulthood. *Journal of Personality, 78*, 1383-1410.

Lodi-Smith, J., & Roberts, B. W. (2012). Concurrent and prospective relationships between social engagement and personality traits in older adulthood. *Psychology and Aging, 27*, 720–727.

Loehlin, J. C. (1992). *Genes and environment in personality development*. Newbury Park, CA: Sage.

Lucas, R. E., Clark, A. E., Georgellis, Y., & Diener, E. (2003). Reexamining adaptation and the set point model of happiness: Reactions to changes in marital status. *Journal of Personality and Social Psychology, 84*, 527-539. doi: 10.1037/0022-3514.84.3.527

Lucas, R. E., & Donnellan, M. B. (2011). Personality development across the life span: Longitudinal analyses with a national sample from Germany. *Journal of Personality and Social Psychology, 101*, 847-861. doi: 10.1037/a0024298

Lüdtke, O., Trautwein, U., & Husemann, N. (2009). Goal and personality trait development in a transitional period: Assessing change and stability in personality development. *Personality and Social Psychology Bulletin, 35*, 428-441.

Lüdtke, O., Roberts, B. W., Trautwein, U., & Nagy, G. (2011). A random walk down university avenue: Life paths, life events, and personality trait change at the transition to university life. *Journal of Personality and Social Psychology, 101*, 620-637. doi: 10.1037/a0023743

Luhmann, M., Hofmann, W., Eid, M., & Lucas, R. E. (2012). Subjective well-being and adaptation to life events: a meta-analysis. *Journal of Personality and Social Psychology, 102*, 592-615. doi: 10.1037/a0025948

Luhmann, M., Orth, U., Specht, J., Kandler, C., & Lucas, R. E. (2014). Studying changes in life circumstances and personality: It's about time. *European Journal of Personality, 28*, 256-266. doi: 10.1002/per.1951

Lutz, W., Ehrlich, T., Rubel, J., Hallwachs, N., Röttger, M.-A., Jorasz, C., . . . Tschitsaz-Stucki, A. (2013). The ups and downs of psychotherapy: Sudden gains and sudden losses identified with session reports. *Psychotherapy Research, 23*, 14-24. doi: 10.1080/10503307.2012.693837

MacLeod, C., Rutherford, E., Campbell, L., Ebsworthy, G., & Holker, L. (2002). Selective attention and emotional vulnerability: Assessing the causal basis of their association through the experimental manipulation of attentional bias. *Journal of Abnormal Psychology, 111*, 107-123. doi: 10.1037/0021-843X.111.1.107

Magidson, J. F., Roberts, B. W., Collado-Rodriguez, A., & Lejuez, C. W. (2014). Theory-driven intervention for changing personality: Expectancy value theory, behavioral activation, and conscientiousness. *Developmental Psychology, 50*, 1442-1450. doi: 10.1037/a0030583

Magnusson, D. (1981). Wanted: A psychology of situations. In D. Magnusson (Ed.), *Toward a psychology of situations: An interactional perspective* (pp. 9-32). Hillsdale, NJ: Lawrence Earlbaum.

Markus, H., & Nurius, P. (1986). Possible selves. *American Psychologist, 41*, 954-969.

Marschall, S. J., & Biddle, S. J. H. (2001). The transtheoretical model of behavior change: A meta-analysis of applications to physical activity and exercise. *Annals of Behavioral Medicine, 23*, 229-246.

Martin, L., Oades, L. G., & Caputi, P. (2014). Intentional personality change coaching: A randomized control trial of client chosen personality facet change using the five-factor model of personality. *International Coaching Psychology Review, 9*, 196-209.

Mathews, A., & MacLeod, C. (2002). Induced processing biases have causal effects on anxiety. *Cognition & Emotion, 16*, 331-354. doi: 10.1080/02699930143000518

Mathews, A., & MacLeod, C. (2005). Cognitive vulnerability to emotional disorders. *Annual Review of Clinical Psychology, 1*, 167-195. doi: 10.1146/annurev.clinpsy.1.102803.143916

Mayer, J. D. (2004). How does psychotherapy influence personality? A theoretical integration. *Journal of Clinical Psychology, 60*, 1291-1315. doi: 10.1002/jclp.20076

McAdams, D. P. (1995). What do we know when we know a person? *Journal of Personality, 63*, 365-396.

McAdams, D. P. (2015). *The art and science of personality development*. New York: Guilford.

McAdams, D. P., & Olson, B. D. (2010). Personality development: Continuity and change over the life course. *Annual Review of Psychology, 61*, 517-542.

McAdams, D. P., & Pals, J. L. (2006). A new Big Five. *American Psychologist, 61*, 204-217.

McAdams, D. P., Ruetzel, K., & Foley, J. M. (1986). Complexity and generativity at midlife: A study of biographical scripts for the future. *Journal of Personality and Social Psychology, 50*, 800-807.

McArdle, J. J., Ferrer-Caja, E., Hamagami, F., & Woodcock, R. W. (2002). Comparative longitudinal structural analyses of the growth and decline of multiple intellectual abilities over the life span. *Developmental Psychology, 38*, 115-142.

McCabe, K. O., & Fleeson, W. (2016). Are traits useful? Explaining trait manifestations as tools in the pursuit of goals. *Journal of Personality and Social Psychology, 110*, 287-301. doi: 10.1037/a0039490

McCrae, R. R. (2003). Expansion, integration, and omission: Some thoughts on a new agenda. *Journal of Gerontology, 58B*, 307-308.

McCrae, R. R., & Costa Jr, P. T. (2008). The Five-Factor Theory of personality. In O. P. John, R. W. Robins, & L. A. Pervin (Eds.), *Handbook of personality theory and research* (pp. 159-181). New York: Guilford Press.

McCrae, R. R., Costa Jr, P. T., Hrebickova, M., Ostendorf, F., Angleitner, A., Avia, M. D., Sanz, J., Sanchez-Bernardos, M. L., Kusdil, M. E., Woodfield, R., Saunders, P. R., & Smith, P. B. (2000). Nature over nurture: Temperament, personality, and life span development. *Journal of Personality & Social Psychology, 78*, 173-186. doi: 10.1037//0022-3514.78.1.173

McCrae, R. R., Costa, P. T., Ňrebíčková, M., Urbánek, T., Martin, T. A., Oryol, V. E., Rukavishnikov, A. A., & Senin, I. G. (2004). Age differences in personality traits across cultures: Self-report and observer perspectives. *European Journal of Personality, 18*, 143-157. doi: 10.1002/per.510

McLean, K. C., Pasupathi, M., & Pals, J. L. (2007). Selves creating stories creating selves: A process model of self-development. *Personality and Social Psychology Review, 11*, 262-278. doi: 10.1177/1088868307301034

Mischel, W., & Shoda, Y. (1995). A cognitive-affective system theory of personality: Reconceptualizing situations, dispositions, dynamics, and invariance in personality structure. *Psychological Review, 102*, 246-268.

Mischel, W., & Shoda, Y. (1998). Reconciling processing dynamics and personality dispositions. *Annual Review of Psychology, 49*, 229-258.

Mõttus, R., Johnson, W., & Deary, I. J. (2012a). Personality traits in old age: Measurement and rank-order stability and some mean-level change. *Psychology and Aging, 27*, 243-249. doi: 10.1037/a0023690

Mõttus, R., Johnson, W., Starr, J. M., & Deary, I. J. (2012b). Correlates of personality trait levels and their changes in very old age: The Lothian Birth Cohort 1921. *Journal of Research in Personality, 46*, 271-278. doi: 10.1016/j.jrp.2012.02.004

Mühlig-Versen, A., Bowen, C. E., & Staudinger, U. M. (2012). Personality plasticity in later adulthood: Contextual and personal resources are needed to increase openness to new experiences. *Psychology and Aging, 27*, 855-866. doi: 10.1037/a0029357

Mund, M., & Neyer, F. J. (2014). Treating personality-relationship transactions with respect: Narrow facets, advanced models, and extended time frames. *Journal of personality and social psychology, 107*, 352- 368. doi: 10.1037/a0036719

Nesselroade, J. R. (2004). Intraindividual variability and short-term change: Commentary. *Gerontology, 50*, 44-47. doi: 10.1159/000074389

Nesselroade, J. R., & Molenaar, P. C. M. (2010). Emphasizing intraindividual variability in the study of development over the lifespan. In W. F. Overton (Ed.), *The handbook of life-span development: Cognition, biology, and methods across the lifespan* (Vol. 1, pp. 30-54). Hoboken, NJ: Wiley.

Neyer, F. J., & Asendorpf, J. B. (2001). Personality-relationship transaction in young adulthood. *Journal of Personality and Social Psychology, 81*, 1190-1204.

Neyer, F. J., & Lehnart, J. (2007). Relationships matter in personality development: Evidence from an 8-year longitudinal study across young adulthood. *Journal of Personality, 75*, 535-568.

Neyer, F. J., Mund, M., Zimmermann, J., & Wrzus, C. (2014). Personality-relationship

transactions revisited. *Journal of Personality, 82*, 539-550. doi: 10.1111/jopy.12063

Nigg, C. R., Burbank, P. M., Padula, C., Dufresne, R., Rossi, J. S., Velicer, W. F., Laforge, R. G., & Prochaska, J. O. (1999). Stages of change: Across ten health risk behaviors for older adults. *The Gerontologist, 39*, 473-482.

Noftle, E. E., & Fleeson, W. (2010). Age differences in Big Five behavior averages and variabilities across the adult life span: Moving beyond retrospective, global summary accounts of personality. *Psychology and Aging, 25*, 95-107. doi: 10.1037/a0018199

Orth, U., Robins, R. W., & Meier, L. L. (2009). Disentangling the effects of low self-esteem and stressful events on depression: Findings from three longitudinal studies. *Journal of Personality and Social Psychology, 97*, 307-321. doi: 10.1037/a0015645

Oyserman, D., Coon, H. M., & Kemmelmeier, M. (2002). Rethinking individualism and collectivism: Evaluation of theoretical assumptions and meta-analyses. *Psychological Bulletin, 128*, 3-72. doi: 10.1037//0033-2909.128.1.3

Pals, J. L. (2006). Narrative identity processing of difficult life experiences: Pathways of personality development and positive self-transformation in adulthood. *Journal of Personality, 74*, 1079-1110. doi: 10.1111/j.1467-6494.2006.00403.x

Paris, R., & Helson, R. (2002). Early mothering experience and personality change. *Journal of Family Psychology, 16*, 172-185. doi: 10.1037//0893-3200.16.2.172

Penton-Voak, I. S., Thomas, J., Gage, S. H., McMurran, M., McDonald, S., & Munafò, M. R. (2013). Increasing recognition of happiness in ambiguous facial expressions reduces anger and aggressive behavior. *Psychological Science, 24*, 688-697. doi: 10.1177/0956797612459657

Piaget, J., & Inhelder, B. (1969). *The psychology of the child*. New York: Basic Books.

Prochaska, J. O., & DiClemente, C. C. (1982). Transtheoretical therapy: Toward a more integrative model of change. *Psychotherapy: Theory, Research and Practice, 19*, 276-288.

Rauthmann, J. F. (2015). Structuring situational information: A road map of the multiple pathways to different situational taxonomies. *European Psychologist*.

Rauthmann, J. F., Gallardo-Pujol, D., Guillaume, E. M., Todd, E., Nave, C. S., Sherman, R. A., Ziegler, M., Jones, A., & Funder, D. C. (2014). The situational eight DIAMONDS: A taxonomy of major dimensions of situation characteristics. *Journal of Personality and Social Psychology, 107*, 677-718. doi: 10.1037/a0037250

Rauthmann, J., Sherman, R. A., & Funder, D. C. (2015). Principles of situation research: Towards a better understanding of psychological situations. *European Journal of Personality, 29*, 363-381. doi: 10.1002/per.1994

Rauthmann, J. F., Sherman, R. A., Nave, C. S., & Funder, D. C. (2015). Personality-driven situation experience, contact, and construal: How people's personality traits predict characteristics of their situations in daily life. *Journal of Research in Personality, 55*, 98-111.

Roberts, B. W. (2006). Personality development and organizational behavior. In B. M. Staw (Ed.), *Research on Organizational Behavior* (Vol. 27, pp. 1-40). New York, NY: Elsevier Science/JAI Press.

Roberts, B. W. (2009). Back to the future: Personality and assessment and personality development. *Journal of Research in Personality, 43*, 137-145. doi: 10.1016/j.jrp.2008.12.015

Roberts, B. W., & Caspi, A. (2003). The cumulative continuity model of personality development: Striking a balance between continuity and change in personality traits across the life course. In R. M. Staudinger, & Lindenberger (Eds.), *Understanding human development: Lifespan psychology in exchange with other disciplines* (pp. 183-214). Dordrecht, NL: Kluwer Academic Publishers.

Roberts, B. W., Caspi, A., & Moffitt, T. E. (2003). Work experiences and personality development in young adulthood. *Journal of Personality and Social Psychology, 84*, 582-593.

Roberts, B. W., & Damian, R. (in press). The principles of personality trait development and their relation to psychopathology. In D. Lynam & D. Samuel (Eds.), *Purdue*

Symposium on Psychological Sciences. Oxford, UK: Oxford University Press.

Roberts, B. W., & DelVecchio, W. F. (2000). The rank-order consistency of personality traits from childhood to old age: A quantitative review of longitudinal studies. *Psychological Bulletin, 126*, 3-25.

Roberts, B. W., & Jackson, J. J. (2008). Sociogenomic personality psychology. *Journal of Personality, 76*, 1523-1544.

Roberts, B. W., & Mroczek, D. (2008). Personality trait change in adulthood. *Current Directions in Psychological Science, 17*, 31-35. doi: 10.1111/j.1467-8721.2008.00543.x

Roberts, B. W., O'Donnell, M., & Robins, R. W. (2004). Goal and personality trait development in emerging adulthood. *Journal of Personality & Social Psychology, 87*, 541-550.

Roberts, B. W., & Robins, R. W. (2004). Person-environment fit and its implications for personality development: A longitudinal study. *Journal of Personality, 72*, 89-110.

Roberts, B. W., Walton, K., Bogg, T., & Caspi, A. (2006a). De-investment in work and non-normative personality trait change in young adulthood. *European Journal of Personality, 20*, 461-474. doi: 10.1002/per.607

Roberts, B. W., Walton, K. E., & Viechtbauer, W. (2006b). Patterns of mean-level change in personality traits across the life course: A meta-analysis of longitudinal studies. *Psychological Bulletin, 132*, 1-25. doi: 10.1037/0033-2909.132.1.1

Roberts, B. W., & Wood, D. (2006). Personality development in the context of the neo-socioanalytic model of personality. In D. Mroczek, & T. D. Little (Eds.), *Handbook of Personality Development* (pp. 11-39). Mahwah, NJ: Lawrance Erlbaum Associates.

Roberts, B. W., Wood, D., & Caspi, A. (2008). The development of personality traits in adulthood. In O. P. John, R. W. Robins, & L. A. Pervin (Eds.), *Handbook of Personality: Theory and research* (3rd ed., pp. 375-398). New York, NY: Guilford Press.

Robins, R. W., & John, O. P. (1997). The quest for self-insight: Theory and research on accuracy and bias in self-perception. In R. Hogan, J. A. Johnson & S. R. Briggs (Eds.), *Handbook of Personality Psychology*. New York: Academic.

Rothman, A. J., Sheeran, P., & Wood, W. (2009). Reflective and automatic processes in the initiation and maintenance of dietary change. *Annals of Behavioral Medicine, 38*, 4-17. doi: 10.1007/s12160-009-9118-3

Ryan, R. M., & Deci, E. L. (2008). Self-determination theory and the role of basic psychological needs in personality and the organization of behavior. In O. John, R. Roberts & L. A. Pervin (Eds.), *Handbook of personality: Theory and research* (pp. 654–678). New York, NY: Guilford Press.

Sadler, P., & Woody, E. (2003). Is who you are who you're talking to? Interpersonal style and complementarily in mixed-sex interactions. *Journal of Personality and Social Psychology, 84*, 80-96. doi: 10.1037/0022-3514.84.1.80

Schmitt, M., Gollwitzer, M., Baumert, A., Blum, G. S., Gschwendner, T., Hofmann, W., & Rothmund, T. (2013). Proposal of a nonlinear interaction of person and situation (NIPS) model. *Frontiers in Psychology, 4*, Article 499. doi: 10.3389/fpsyg.2013.00499

Schmitt, M., Perugini, M., Blum, G., Baumert, A., & Johnson, W. (2016). Integrating personality structure and processes. *Manuscript in preparation.*

Schröder-Abé, M., Rudolph, A., & Schütz, A. (2007). High implicit self-esteem is not necessarily advantageous: Discrepancies between explicit and implicit self-esteem and their relationship with anger expression and psychological health. *European Journal of Personality, 21*, 319-339.

See, J., MacLeod, C., & Bridle, R. (2009). The reduction of anxiety vulnerability through the modification of attentional bias: A real-world study using a home-based cognitive bias modification procedure. *Journal of Abnormal Psychology, 118*, 65-75. doi: 10.1037/a0014377

Seger, C. A. (1994). Implicit learning. *Psychological Bulletin, 115*, 163. doi: 10.1037/0033-2909.115.2.163

Sheldon, K. M., Ryan, R. M., Deci, E. L., & Kasser, T. (2004). The independent effects of goal contents and motives on well-being: It's both what you pursue and why you pursue it.

Personality and Social Psychology Bulletin, 30, 475-486.

Sheldon, K. M., & Schüler, J. (2011). Wanting, having, and needing: Integrating motive disposition theory and self-determination theory. *Journal of Personality and Social Psychology, 101*, 1106-1123. doi: 10.1037/a0024952

Sherman, R. A., Rauthmann, J. F., Brown, N. A., Serfass, D. G., Jones, A. B., & John, F. (2015). The independent effects of personality and situations on real-time expressions of behavior and emotion. *Journal of Personality and Social Psychology*. doi: 10.1037/pspp0000036

Slavich, G. M., & Irwin, M. R. (2014). From stress to inflammation and major depressive disorder: A social signal transduction theory of depression. *Psychological Bulletin, 140*, 774-815. doi: 10.1037/a0035302

Sliwinski, M. J. (2008). Measurement-burst designs for social health research. *Social and Personality Psychology Compass, 2*, 245-261. doi: 10.1111/j.1751-9004.2007.00043.x

Small, B. J., Hertzog, C., Hultsch, D. F., & Dixon, R. A. (2003). Stability and change in adult personality over 6 years: Findings from the Victoria Longitudinal Study. *Journals of Gerontology, 58B*, 166-176.

Smith, J., & Baltes, P. B. (1997). Profiles of psychological functioning in the old and oldest old. *Psychology and Aging, 12*, 458-472. doi: 10.1037/0882-7974.12.3.458

Snyder, M., & Ickes,W. (1985). Personality and social behavior. In G. Lindzey, & E. Aronson (Eds.), *The handbook of social psychology* (pp. 883-947). New York: Random House.

Soto, C. J., & John, O. P. (2009). Ten facet scales for the Big Five Inventory: Convergence with NEO PI-R facets, self-peer agreement, and discriminant validity. *Journal of Research in Personality, 43*, 84-90. doi: 10.1016/j.jrp.2008.10.002

Soto, C. J., & John, O. P. (2012). Development of Big Five domains and facets in adulthood: Mean-level age trends and broadly versus narrowly acting mechanisms. *Journal of Personality, 80*, 881-914. doi: 10.1111/j.1467-6494.2011.00752.x

Spalding, L. R., & Hardin, C. D. (1999). Unconscious unease and self-handicapping: Behavioral consequences of individual differences in implicit and explicit self-esteem. *Psychological Science, 10*, 535-539.

Specht, J., Bleidorn, W., Denissen, J. J., Hennecke, M., Hutteman, R., Kandler, C., ... & Zimmermann, J. (2014). What drives adult personality development? A comparison of theoretical perspectives and empirical evidence. *European Journal of Personality, 28*, 216-230.

Specht, J., Egloff, B., & Schmukle, S. C. (2011). Stability and change of personality across the life course: The impact of age and major life events on mean-level and rank-order stability of the Big Five. *Journal of Personality and Social Psychology, 101*, 462-482.

Staudinger, U. M. (2001). Life reflection. *Review of General Psychology, 5*, 148-160.

Staudinger, U. M., & Kunzmann, U. (2005). Positive adult personality development: Adjustment and/or growth? *European Psychologist, 10*, 320-329. doi: 10.1027/1016-9040.10.4.320

Steinhagen-Thiessen, E., & Borchelt, M. (1999). Morbidity, medication, and functional limitations in very old age. In P. B. Baltes, & K. U. Mayer (Eds.), *The Berlin Aging Study: Aging from 70 to 100* (pp. 131-166). Cambridge: Cambridge University Press.

Strack, F., & Deutsch, R. (2004). Reflective and impulsive determinants of social behavior. *Personality & Social Psychology Review, 8*, 220-247.

Swann Jr, W. B. (1983). Self-verification: Bringing social reality into harmony with the self. In J. Suls, & A. G. Greenwald (Eds). *Social psychological perspectives on the self* (p.33-66). Hillsdale, NJ: Erlbaum.

Tackett, J. L., Slobodskaya, H. R., Mar, R. A., Deal, J., Halverson, C. F., Baker, S. R., … Besevegis, E. (2012). The hierarchical structure of childhood personality in five countries: Continuity from early childhood to early adolescence. *Journal of Personality, 80*, 847–879. DOI:10.1111/j.1467-6494.2011.00748.x

Tang, T. Z., & DeRubeis, R. J. (1999). Sudden gains and critical sessions in cognitive-behavioral therapy for depression. *Journal of Consulting and Clinical Psychology, 67*, 894-904.

Terracciano, A., Costa, P. T., & McCrae, R. R. (2006). Personality plasticity after age 30. *Personality and Social Psychology Bulletin, 32*, 999-1009. doi: 10.1177/0146167206288599

Terracciano, A., McCrae, R. R., Brant, L. J., & Costa Jr, P. T. (2005). Hierarchical linear modeling analyses of the NEO-PI-R scales in the Baltimore Longitudinal Study of Aging. *Psychology and Aging, 20*, 493-506. doi: 10.1037/0882-7974.20.3.493

Trzesniewski, K. H., Donnellan, M. B., & Robins, R. W. (2003). Stability of self-esteem across the life span. *Journal of Personality and Social Psychology, 84*, 205-220. doi: 10.1037/0022-3514.84.1.205

Tucker-Drob, E. M., Briley, D. A., & Harden, K. P. (2013). Genetic and environmental influences on cognition across development and context. *Current Directions in Psychological Science, 22*, 349-355. doi: 10.1177/0963721413485087

Tucker-Drob, E. M., & Briley, D. A. (2014). Continuity of genetic and environmental influences on cognition across the life span: A meta-analysis of longitudinal twin and adoption studies. *Psychological Bulletin, 140*, 949–979. doi: 10.1037/a0035893

van Aken, M. A. G., Denissen, J. J. A., Branje, S. J. T., Dubas, J. S., & Goossens, L. (2006). Midlife concerns and short-term personality change in middle adulthood. *European Journal of Personality, 20*, 497-513.

Vukman, K. B. (2005). Developmental differences in metacognition and their connections with cognitive development in adulthood. *Journal of Adult Development, 12*, 211-221. doi: doi: 10.1007/s10804-005-7089-6

Wagerman, S. A., & Funder, D. C. (2009). Situations. In P. J. Corr & G. Matthews (Eds.), *Cambridge handbook of personality* (pp. 27–42). Cambridge, England: Cambridge University Press.

Wagner, J., Becker, M., Trautwein, U., & Lüdtke, O. (2015). The first partnership experience and personality development: A propensity score matching study in young adulthood. *Social Psychological and Personality Science*. doi: 10.1177/1948550614566092

Wagner, J., Lüdtke, O., Jonkmann, K., & Trautwein, U. (2013). Cherish yourself: Longitudinal patterns and conditions of self-esteem change in the transition to young adulthood. *Journal of Personality and Social Psychology, 104*, 148-. doi: 10.1037/a0029680

Wagner, J., Ram, N., Smith, J., & Gerstorf, D. (2015). Personality trait development at the end of life: Antecedents and correlates of mean-level trajectories *Journal of Personality and Social Psychology*. doi: 10.1037/pspp0000071

Ward, C., Leong, C.-H., & Low, M. (2004). Personality and sojourner adjustment an exploration of the big five and the cultural fit proposition. *Journal of Cross-Cultural Psychology, 35*, 137-151. doi: 10.1177/0022022103260719

Watson, J. B. (1913). Psychology as the behaviorist views it. *Psychological Review, 20*, 158-177.

Webb, T. L., & Sheeran, P. (2006). Does changing behavioral intentions engender behavior change? A meta-analysis of the experimental evidence. *Psychological Bulletin, 132*, 249-268. doi: 10.1037/0033-2909.132.2.249

Williams, E. F., & Gilovich, T. (2008). Conceptions of the self and others across time. *Personality and Social Psychology Bulletin, 34*, 1037-1046.

Wilson, T. D. (2004). *Strangers to ourselves*: Harvard University Press.

Wilson, T. D., & Dunn, E. W. (2004). Self-knowledge: Its limits, value, and potential for improvement. *Annual Review of Psychology, 55*, 493-518.

Wood, A. M., Brown, G. D. A., Maltby, J., & Watkinson, P. (2012). How are personality judgments made? A cognitive model of reference group effects, personality scale responses, and behavioral reactions. *Journal of Personality, 80*, 1275-1311. doi: 10.1111/j.1467-6494.2012.00763.x

Wood, D., & Denissen, J. J. A. (2015). A functional perspective on personality trait development. In N. R. Branscombe & K. Reynolds (Eds.), *Psychology of change: Life contexts, experiences, and identities* (pp. 97-115). New York: Psychology Press.

Wood, D., Gardner, M. H., & Harms, P. (2015). How functionalist and process approaches to behavior can explain trait covariation. *Psychological Review, 122*, 84-111. doi: 10.1037/a0038423

Wood, W., & Neal, D. T. (2007). A new look at habits and the habit-goal interface. *Psychological Review, 114*, 843-863. doi: 10.1037/0033-295X.114.4.843

Wood, D., & Roberts, B. W. (2006). The effect of age and role information on expectations for Big Five personality traits. *Personality and Social Psychology Bulletin, 32*, 1482-1496.

Wortman, J., Lucas, R. E., & Donnellan, M. B. (2012). Stability and change in the Big Five personality domains: Evidence from a longitudinal study of Australians. *Psychology and Aging, 27*, 867-874. doi: 10.1037/a0029322

Wrzus, C., Hänel, M., Wagner, J., & Neyer, F. J. (2013a). Social network change and life events across the lifespan: A meta-analysis. *Psychological Bulletin, 139*, 53-80. doi: 10.1037/a0028601

Wrzus, C., & Lang, F. R. (2010). Entwicklung der Persönlichkeit [Personality development]. In F. R. Lang, M. Martin & M. Pinquart (Eds.), *Entwicklungspsychologie - Erwachsenenalter* (pp. 141-159). Göttingen: Hogrefe.

Wrzus, C., Luong, G., Wagner, G. G., & Riediger, M. (2016). Increases in short-term affective responses to daily hassles predicts increases in neuroticism over six years. *Manuscript submitted for publication*.

Wrzus, C., Müller, V., Wagner, G. G., Lindenberger, U., & Riediger, M. (2013b). Affective and cardiovascular responding to unpleasant events from adolescence to old age: Complexity of events matters. *Developmental Psychology, 49*, 384-397. doi: 10.1037/a0028325

Wrzus, C., & Neyer, F. J. (2016). Co-development of personality and friendships across the lifespan: An empirical review on selection and socialization. *Manuscript submitted for publication.*

Wrzus, C., Wagner, G. G., & Riediger, M. (2016). Personality-situation transactions from adolescence to old age. *Journal of Personality and Social Psychology, 110*, 782-799. doi: 10.1037/pspp0000054

Zimmermann, J., & Neyer, F. J. (2013). Do we become a different person when hitting the road? Personality development of sojourners. *Journal of Personality & Social Psychology, 105*, 515-530. doi: 10.1037/a0033019

Footnotes

[1] A psychological process is a series of connected parts, such as situations, thoughts, emotions, behavior (Schmitt et al., 2016). A process implies a passing of time.

[2] Short-term processes occur within seconds, minutes or hours, and long-term processes usually cover years and decades (see Gerstorf, Hoppmann, & Ram, 2014).

[3] Expectancy was not assessed in the study. We discuss later that expectancies may be an optional component during the TESSERA sequence as situations and their relevant psychological properties can elicit states directly (Figure 1).

[4] For the clarity of the framework, we propose that one specific state follows a certain triggering situation (Figure 1). Mischel and Shoda (1995) similarly described in the CAPS model that specific situations and their psychologically relevant characteristics elicit a distinct state. Very likely though, in reality different aspects of states, that is, thoughts, feelings, and behaviors co-occur. It will be a methodological matter to delineate how such co-occurring states precede, accompany, or follow each other.

[5] People with worse health are at greater risk of experiencing illnesses or unemployment (Adler & Newman, 2002).

[6] Adaptiveness of a behavior or personality characteristic refers here to a benefit for the momentary situation (e.g., state well-being) or for a period of a single life (e.g., maintaining health and well-being). This is distinct from the evolutionary meaning, where adaptiveness refers to contributing to the survival and reproduction of a group or species.

[7] To assess life narratives, people are asked to tell about difficult times in their life. Except for questions about how old people were, when they experienced the difficulties and how long difficulties lasted, no further structured questions are posed. Still many life narratives possess the content of stories: what was the setting = triggering situation; what happened = behaviors, thoughts, feelings; how did they react and what did other people do = reactions (McAdams, 1995; Pals, 2006; own data available upon request).

[8] We focus on adolescence and adulthood because childhood personality development may pose some exceptions. For example, the personality structure is less established during childhood (Tackett et al., 2012) and available metacognitive competencies may restrict processes such as expectancy formation and reflection (Flavell, 2000).

[9] Terracciano and colleagues (2006) reported similar rank-order correlations for younger, middle-aged and older people. The high educational level and socio-economic status of the sample might account for discrepancies with finding from larger and more diverse samples (Lucas & Donellan, 2011; Wortman et al., 2012) and with meta-analytic results (Ardelt, 2000) that showed decreases in rank-order stability in late adulthood.

Table 1

Overview on Central Propositions of the TESSERA Framework

TESSERA components	Repeated equences of Triggering situations, Expectancies, States/ State expressions, and Reactions lead to personality change and continuity (i.e., personality development) over time through associative and reflective processes (see next row).
Associative & reflective processes	The amount and direction of change depends on the associative and reflective processes linking short-term TESSERA sequences to long-term personality development. Reflective processes include, for example, self-reflection, accomodation, assimilation, life reflection, self-narration. Personality development can also occur without reflection through associative processes (e.g., implicit learning, reinforcement learning, habit formation), when similar TESSERA sequences occur repeatedly.
Moderators	TESSERA components can vary in being positive—negative, internal—external, automatic—deliberate, and with respect to resources and constraints of the individual and the environment.
Age-sensitivity	TESSERA components and processes vary with age due to physical, cognitive, social, and societal changes related to age.
Genericness	The framework, its short-, and long-term processes, applies to different levels of personality, that is, behavioral dispositions, motivational orientations, and narrative descriptions.

Note. Table 1 summarizes the central propositions of the framework, which are detailed throughout the article.

Table 2

Processes of Personality Development described in Personality, Developmental, Social or Clinical psychology

Process	**Author/source**	**Brief description**	**Time scale**[a]	**Relation to TESSERA**
Cumulative continuity	Roberts & Caspi, 2003; Roberts & Wood, 2006	increasing stability of individual differences in personality due to genes, environment, identity, and also personality maturity	long-term	set of processes /principles; repeated TESSERA sequences can contribute to cumulative continuity
Genetic influences	Caspi & Roberts, 2001	genes partly shape personality differences and may support continuity by influencing biological processes how stimuli are perceived and reacted to incl. selection of environment	long-term	moderators that can influence triggering situations, expectancies, states, state expressions, reactions, and reflections as well as their relationship
Environment	Caspi & Roberts, 2001; Roberts & Wood, 2006	environmental factors (e.g., people, society, surroundings) provide constraints to experiences	medium-long-term	provide triggering situations & constraints for states
Social roles & social investment	Roberts & Wood, 2006	social roles (e.g., supervisor, mother) provide proximate environment for personality continuity and change	medium-long-term	provide triggering situations and expectancies
Identity development	Roberts & Caspi, 2003; Greve et al., 2005	forming and maintaining a coherent and relatively stable view on oneself and one's abilities, preferences, characteristics, experiences	medium-term	set of reflective processes
Life reflection	Staudinger, 2001	remembering and analyzing/ evaluating past experiences to achieve insight	medium-term	reflective process
Self-narration & subprocesses	Hooker & McAdams 2003b; King, 2001; Pals, 2006	remembering and storytelling of previous (life) events	short- to medium-term	reflective process
Generalization	Back et al., 2009 Krampen, 1988	reflective processes "condense" into explicit self-representations	short-term	repeated reflective process

Table 2 (*continued*)

Process	Author/source	Brief description	Time scale[a]	Relation to TESSERA
Person-Environ-ment transactions/ Corresponsive principle	Buss, 1987; Caspi & Roberts, 2001; Roberts & Wood, 2006	people select and create different environments in correspondence with their personality (selection, manipulation), react differently to the same environment (reactive), or evoke different reactions from the same environment	short-term	processes how personality-(in)congruent triggering situations and states can occur
Self-regulation & subprocesses	Hooker & Mc Adams, 2003b	(control) processes that change one's thoughts and actions to facilitate goal pursuit	short-term	expectancies, reactions
Self-reflection	Bem, 1972; Caspi & Roberts, 2001	achieving information from watching & thinking about ones behavior, thought, and feelings	short-term	reflective process
Accommodation & assimilation	Brandtstädter, 1989; King, 2001; Piaget & Inhelder, 1969	changing ones view of oneself or the (social) environment; integrating experiences into existing views of oneself	short-term	reflective process
Positive reframing	King, 2001	seeing positive aspects about an event	short-term	reflective process
Implicit learning	Seger, 1994	repeatedly co-occurring cognitive activation of concepts	short-term	associative process
Habit formation/change	Rothmann et al., 2009; Wood & Neal, 2007	implicit learning of repeated behavior	short-term	associative process
Feedback	Caspi & Roberts, 2001	receiving information from others on one's behavior, thought, and feelings	short-term	reaction, part of the associative process
Reinforcement	Caspi & Roberts, 2001	changing & maintaining one's behavior when experiencing pleasant or unpleasant reactions (from others) when showing the behavior	short-term	reaction, part of the associative process
Model learning	Bandura,1986; Caspi & Roberts, 2001	watching others' behavior and expression of thought and feeling	short-term	trigger
States	Hooker & Mc Adams, 2003b	varying feelings, behaviors, and thoughts	short-term	state, not a process (McCrae, 2003)

Table note. [a] Short-term processes occur within seconds, minutes or hours, medium-term processes last weeks to months, and long-term processes usually cover years and decades (see Gerstorf, Hoppmann, & Ram, 2014).

Table 3

Overview on Patterns of Personality Development across the Adult Lifespan viewed from within the TESSERA Framework

	Young adulthood	**Middle adulthood**	**Late adulthood**
Main developmental pattern	Increasing Maturity	Continuity	Plasticity and diversification, i.e., growth, stability, or decline among different people
Central empirical findings	Increases in emotional stability, agreeableness, conscientiousness (e.g., Lucas & Donnellan, 2011; Roberts et al., 2006b)	High rank-order correlations and relatively little mean-level change (e.g., Lucas & Donnellan, 2011; Roberts & delVecchio, 2000; Roberts et al., 2006b)	Lower rank-order correlations compared to middle adulthood; differential mean-level changes (e.g., Jackson et al., 2012; Lucas & Donnellan, 2011; Mõttus et al., 2012a)
TESSERA framework explanations	New triggering situations and expectancies arise from the multitude of life transitions and changes in social roles.	Relatively stable environments and social roles provide few new triggering situations. Personality-situation transactions and identity development further contribute to consolidation of personality.	Diffuse expectancies exist for this life period. Varying experiences related to health and cognitive decline provide different triggering situations and constraints for growth and stability.

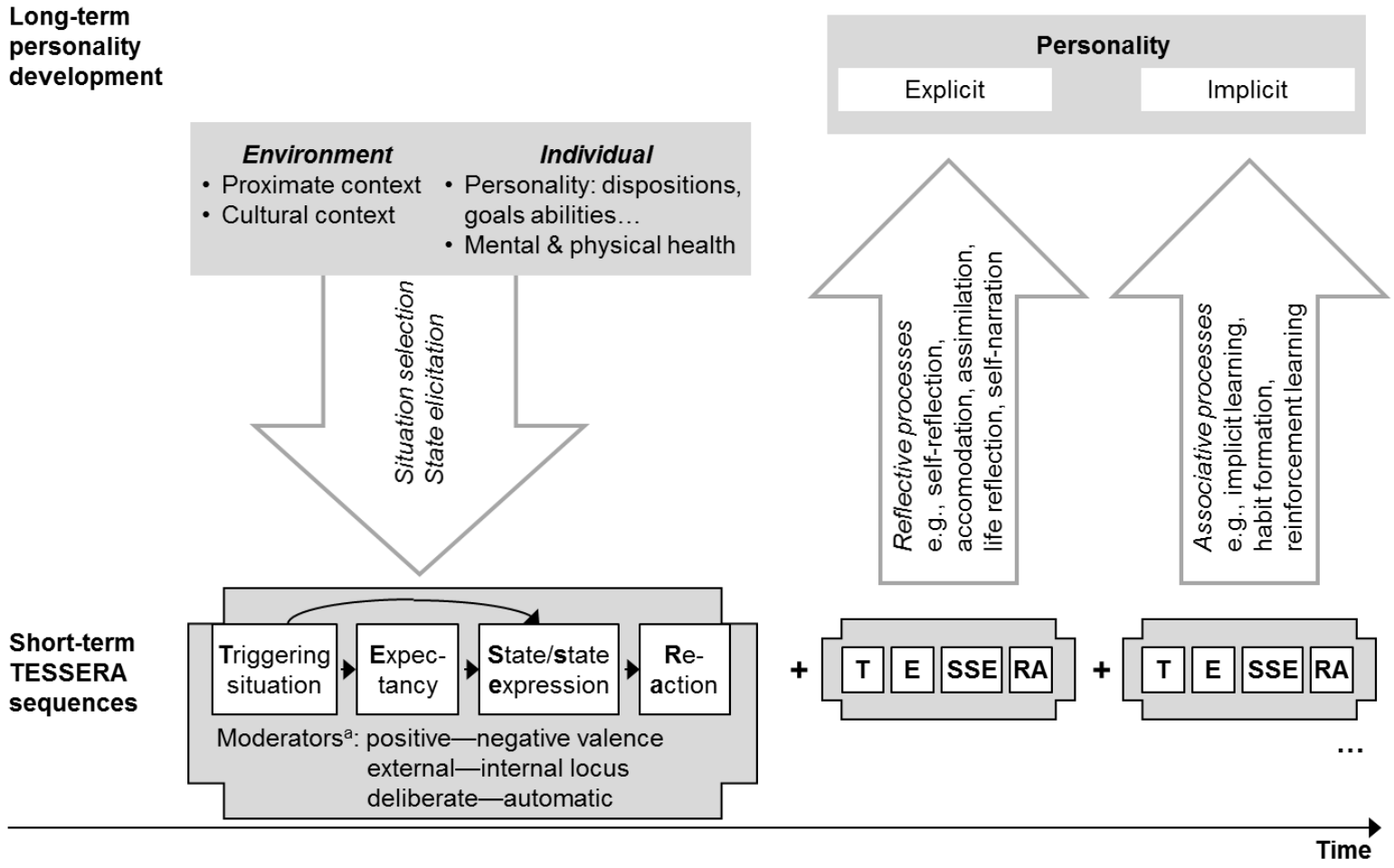

Figure 1. TESSERA framework of adult personality development processes.
Note. We postulate that long-term personality development occurs due to repeated short-term processes. These short-term processes can be generalized as sequences of Triggering situations, Expectancies, States, State expressions, and Reactions (TESSERA). Associative and reflective processes transform repeated TESSERA sequences into long-term personality development. See main text for detailed explanations.

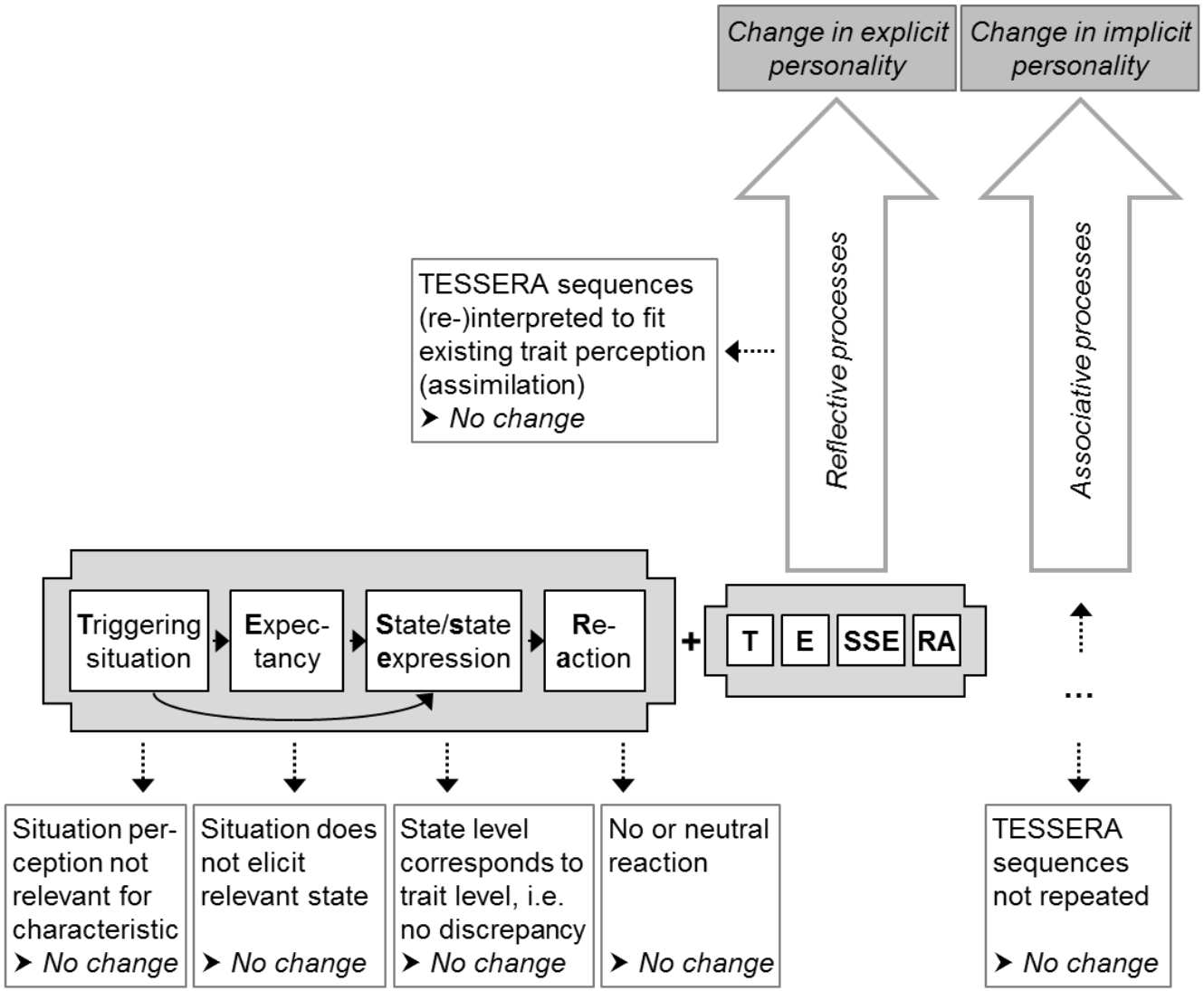

Figure 2. Conditions under which TESSERA sequences, reflective and associative processes do (not) lead to changes in personality characteristics.

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